

前言

1. 阅读预测题的来源是什么？阅读预测题目是如何筛选的？

阅读预测题目均来自于真实考试，由教研组老师们结合回忆信息精心整理。老师们参考了过去几年的雅思阅读考试趋势，并对 2025 年可能出现的话题和题型进行了深入分析，从而挑选并整理出这些题目。

2. 阅读预测题目该如何使用呢？

拿到阅读预测题目后，建议进行专题和模拟练习，特别是针对自身的弱点进行专项训练。此外，对于题目中涉及的词汇，包括同义词替换和话题相关词汇，也需要进行深入学习和整理。在考试前，这些内容应作为复习的重点。这样的备考策略将有助于提高阅读理解能力和考试表现。

3. 如果参加机考，阅读预测的题目是否适用呢？

雅思阅读纸笔考试和机考的题库是一致的，因此机考的同学们也可以使用本题库进行练习。

4. 阅读大范围预测与小范围预测的区别是什么？该如何选择呢？

阅读大范围预测包含 45 篇题目，每个 passage 有 15 篇练习；阅读小范围预测包含 30 篇题目，每个 passage 有 10 篇练习。如果距离考试不到或只有一个月的时间，备考时间紧任务重，可以选择使用小范围题库重点复习。如果在当季参加考试，备考时间充分，可以选择使用大范围题库全面覆盖知识点，不留死角。

5. 阅读备考的重点是什么？

根据 2024 年的考试趋势，雅思阅读部分的填空题、判断题和配对题占比较大，因此，需要进行针对性的训练。主要考察的话题包括动植物、科技、历史、艺术与文化以及心理学，这些

领域的词汇积累是非常重要的。同学们应根据自身的学习状况，制定个性化的复习计划，及时反思并分析错误题目，深入理解并整理同义词替换等知识点。在考试前，通过模拟真题进行练习，以熟悉考试流程和环境，使自己能够在高度集中的状态下应对雅思考试。

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Passage 1

Satellite Technology

The space-age began with the launch of the Russian artificial satellite Sputnik in 1957 and developed further with the race to the moon between the United States and Russia. This rivalry was characterized by advanced technology and huge budgets. In this process, there were spectacular successes, some failures, but also many spin-offs. Europe, Japan, China, and India quickly joined this space club of the superpowers. With the advent of relatively low-cost high-performance mini-satellites and launchers, the acquisition of indigenous space capabilities by smaller nations in Asia has become possible. How, in what manner, and for what purpose will these capabilities be realized?

A Rocket technology has progressed considerably since the days of ‘fire arrows’ (bamboo poles filled with gunpowder) first used in China around 500 BC, and, during the Sung Dynasty, to repel Mongol invaders at the battle of Kaifeng (Kai-fung fu) in AD 1232. These ancient rockets stand in stark contrast to the present-day Chinese rocket launch vehicles, called the ‘Long March’, intended to place a Chinese astronaut in space by 2005 and, perhaps, to achieve a Chinese moon landing by the end of the decade.

B In the last decade, there has been a dramatic growth in space activities in Asia both in the utilization of space-based services and the production of satellites and launchers. This rapid expansion has led many commentators and analysts to predict that Asia will become a world space power. The space age has had dramatic effects worldwide with direct developments in space technology influencing telecommunications, meteorological forecasting, earth resource and environmental monitoring, and disaster mitigation (flood, forest fires, and oil spills). Asian nations have been particularly eager to embrace these developments.

C New and innovative uses for satellites are constantly being explored with potential revolutionary effects, such as in the field of health and telemedicine, distance education, crime prevention (piracy on the high seas), food and agricultural planning and production (rice crop monitoring). Space in Asia is very much influenced by the competitive commercial space sector, the emergence of low-cost mini-satellites, and the globalization of industrial and financial markets. It is not evident how Asian space will develop in the coming decades in the face of these trends. It is, however, important to understand and assess the factors and forces that shape Asian space activities and development in determining its possible consequences for the region.

D At present, three Asian nations, Japan, China, and India, have comprehensive end-to-end space capabilities and possess a complete space infrastructure: space technology, satellite manufacturing, rockets, and spaceports. Already self-sufficient in terms of satellite design and manufacturing, South Korea is currently attempting to join its ranks with its plans to develop a launch site and spaceport. Additionally, nations in Southeast Asia as well as those bordering the Indian subcontinent (Nepal, Pakistan, and Bangladesh), have, or are starting to develop indigenous space programmes. The Association of Southeast Asian Nations (ASEAN) has, in varying degrees, embraced space applications using foreign technology and over the past five years or so its space activities have been expanding. Southeast Asia is predicted to become the largest and fastest-growing market for commercial space products and applications, driven by telecommunications (mobile and fixed services), the Internet, and remote sensing applications. In the development of this technology, many non-technical factors, such as economics, politics, culture, and history, interact and play important roles, which in turn affect Asian technology.

E Asia and Southeast Asia, in particular, suffer from a long list of recurrent large-scale environmental problems

including storms and flooding, forest fires and deforestation, and crop failures. Thus the space application that has attracted the most attention in this region is remote sensing. Remote sensing satellites equipped with instruments to take photographs of the ground at different wavelengths provide essential information for natural resource accounting, environmental management, disaster prevention and monitoring, land-use mapping, and sustainable development planning. Progress in these applications has been rapid and impressive. ASEAN members, unlike Japan, China, and India, do not have their own remote sensing satellites, however, most of its member nations have facilities to receive, process, and interpret such data from American and European satellites. In particular, Thailand, Malaysia, and Singapore have world-class remote sensing processing facilities and research programmes. ASEAN has plans to develop (and launch) its own satellites and in particular remote sensing satellites. Japan is regarded as the dominant space power in Asia and its record of successes and quality of technologies are equal to those of the West. In view of the technological challenges and high risks involved in space activities, a very long, and expensive, learning curve has been followed to obtain those successes achieved. Japan, satellite manufacturing was based on the old and traditional defense and military procurement methodologies as practiced in the US and Europe.

F In recent years there have been fundamental changes in the way satellites are designed and built to drastically reduce costs. The emergence of 'small satellites' and their quick adoption by Asian countries as a way to develop low-cost satellite technology and rapidly establish a space capability has given these countries the possibility to shorten their learning curve by a decade or more. The global increase of technology transfer mechanisms and the use of readily available commercial technology to replace costly space and military-standard components may very well result in a highly competitive Asian satellite manufacturing industry.

G The laws of physics are the same in Tokyo as in Toulouse, and the principles of electronics and mechanics know no political or cultural boundaries. However, no such immutability applies to engineering practices and management; they are very much influenced by education, culture, and history. These factors, in turn, have an effect on costs, lead times, product designs and, eventually, international sales. Many Asian nations are sending their engineers to be trained in the West. Highly experienced, they return to work in the growing Asian space industry. Will this acquisition of technical expertise, coupled perhaps with the world-renowned Japanese manufacturing and management techniques, be applied to build world-class satellites and reduce costs?

Questions 1-5

The reading passage has seven paragraphs, **A-G**

Choose the correct heading for paragraphs **A-G** from the list below.

Write the correct number, i-ix, in boxes 1-5 on your answer sheet.

List of Headings

- i Western countries provide essential assistance
- ii Unbalanced development for an essential space technology
- iii Innovative application compelled by competition
- iv An ancient invention which is related to the future
- v Military purpose of the satellite
- vi Rockets for application in ancient China
- vii Space development in Asia in the past
- viii Non-technology factors count
- ix competitive edge gained by more economically feasible satellite

Paragraph D Example: Current space technology development in Asia

1..... Paragraph **A**

- 2..... Paragraph B
 3..... Paragraph C
 4..... Paragraph E
 5..... Paragraph F

Questions 6-9

Match the following reasons for each question according to the information given in the passage

Write the correct letter A-F, in boxes 6-9 on your answer sheet.

- A Because it helps administrate the crops.
 B Because there are some unapproachable areas. C Because the economic level in that area is low.
 D Because there are influences from some other social factors.
 E Because it can be used for non-peaceful purposes.
 F Because disasters such as bushfires happened in Southeast Asia.
- 6..... Why remote-photographic technology is used to resolve environmental problems?
 7..... Why satellite technology is used in the medical area?
 8..... Why Asian country's satellite technology is limited for development?
 9..... Why satellite technology is deployed in an agricultural area?

Questions 10-13

Do the following statements agree with the information given in Reading Passage?

In boxes **10-13** on your answer sheet, write

TRUE if the statement is true

FALSE if the statement is false

NOT GIVEN if the information is not given in the passage

- 10..... Ancient China had already deployed rockets for a military purpose as early as 500 years ago.
 11..... Space technology has enhanced the literacy of Asia.
 12..... photos taken by satellites with certain technology help predict some natural catastrophes prevention and surveillance.
 13..... commercial competition constitutes a boosting factor to Asian technology development.

Mechanisms of Linguistic Change

A The changes that have caused the most disagreement are those in pronunciation. We have various sources of evidence for the pronunciations of earlier times, such as the spellings, the treatment of words borrowed from other languages or borrowed by them, the descriptions of contemporary grammarians and spelling reformers, and the modern pronunciations in all the languages and dialects concerned. From the middle of the sixteenth century, there are in England writers who attempt to describe the position of the speech-organs for the production of English phonemes, and who invent what are in effect systems of phonetic symbols. These various kinds of evidence, combined with a knowledge of the mechanisms of speech production, can often give us a very good idea of the pronunciation of an earlier age, though absolute certainty is never possible.

B When we study the pronunciation of a language over any period of a few generations or more, we find there are always large-scale regularities in the changes: for example, over a certain period of time, just about all the long [a:] vowels in a language may change into long [e:] vowels, or all the [b] consonants in a certain position (for example at the end of a word) may change into [p] consonants. Such regular changes are often called sound laws. There are no universal sound laws (even though sound laws often reflect universal tendencies), but simply particular sound laws for one given language (or dialect) at one given period

C It is also possible that fashion plays a part in the process of change. It certainly plays a part in the spread of change: one person imitates another, and people with the most prestige are most likely to be imitated, so that a change that takes place in one social group may be imitated (more or less accurately) by speakers in another group. When a social group goes up or down in the world, its pronunciation of Russian, which had formerly been considered desirable, became on the contrary an undesirable kind of accent to have, so people tried to disguise it. Some of the changes in accepted English pronunciation in the seventeenth and eighteenth centuries have been shown to consist in the replacement of one style of pronunciation by another style already existing, and it is likely that such substitutions were a result of the great social changes of the period: the increased power and wealth of the middle classes, and their steady infiltration upwards into the ranks of the landed gentry, probably carried elements of middle-class pronunciation into upper-class speech.

D A less specific variant of the argument is that the imitation of children is imperfect: they copy their parents' speech but never reproduce it exactly. This is true, but it is also true that such deviations from adult speech are usually corrected in later childhood. Perhaps it is more significant that even adults show a certain amount of random variation in their pronunciation of a given phoneme, even if the phonetic context is kept unchanged. This, however, cannot explain changes in pronunciation unless it can be shown that there is some systematic trend in the failures of imitation: if they are merely random deviations they will cancel one another out and there will be no net change in the language.

E One such force which is often invoked is the principle of ease or minimization of effort. The change from fussy to fuzzy would be an example of assimilation, which is a very common kind of change. Assimilation is the changing of a sound under the influence of a neighbouring one. For example, the word scant was once skart, but the /m/ has been changed to /n/ under the influence of the following /t/. Greater efficiency has hereby been achieved, because /n/ and /t/ are articulated in the same place (with the tip of the tongue against the teeth ridge), whereas /m/ is articulated elsewhere (with the two lips). So the place of articulation of the nasal consonant has been changed to conform with that of the following plosive. A more recent example of the same kind of thing is the common pronunciation of football as footbal.

F Assimilation is not the only way in which we change our pronunciation in order to increase efficiency. It is very common for consonants to be lost at the end of a word: in Middle English, word-final [-n] was often lost in unstressed syllables, so that baken 'to bake' changed from ['ba:kan] to ['ba:kɜ], and later to [ba:k]. Consonant clusters are often simplified. At one time there was a [t] in words like castle and Christmas, and an initial [k] in words like knight and know. Sometimes a whole syllable is dropped out when two successive syllables begin with the same consonant (haplology): a recent example is temporary, which in Britain is often pronounced as if it were temporary.

Questions 1-4

Complete the summary below.

Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

Write your answers in boxes **1-4** on your answer sheet.

The pronunciation of living languages undergo changes throughout thousands of years. Large-scale regular Changes are usually called **1**..... There are three reasons for these changes. Firstly, the influence of one language on another; when one person imitates another pronunciation (the most prestigious), the imitation always partly involves factor of **2**..... Secondly, the imitation of children from adults' language sometimes are **3**....., and may also contribute to this change if there are insignificant deviations though later they may be

corrected Finally, for those random variations in pronunciation, the deeper evidence lies in the 4..... or minimization of effort.

Questions 5-11

Do the following statements agree with the information given in Reading Passage?

In boxes **5-11** on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

5..... It is impossible for modern people to find the pronunciation of words from an earlier age

6..... The great change of language in Russian history is related to the rising status and fortune of the middle classes.

7..... All the children learn speeches from adults while they assume that certain language is difficult to imitate exactly.

8..... Pronunciation with causal inaccuracy will not exert a big influence on language changes.

9..... The word 'scant' can be pronounced more easily than 'skamt'

10..... The [g] in gnat not being pronounced will not be spelt out in the future.

11..... The sound of 'temporary' cannot wholly present its spelling.

Questions 12-14

Look at the following sentences and the list of statements below.

Match each statement with the correct sentence, **A-D**.

Write the correct letter, **A-D**, in boxes **12-14** on your answer sheet

A Since the speakers can pronounce it with less effort

B Assimilation of a sound under the influence of a neighbouring one

C It is a trend for changes in pronunciation on a large scale in a given period

D Because the speaker can pronounce [n] and [t] both at the same time

12..... As a consequence, 'b' will be pronounced as

13..... The pronunciation of [mt] changed to [nt]

14..... The omit of 'f in the sound of Christmas

Animal Minds: Parrot Alex

A In 1977 Irene Pepperberg, a recent graduate of Harvard University did something very bold. At a time when animals still were considered automatons, she set out to find what was on another creature's mind by talking to it. She brought a one-year-old African gray parrot she named Alex into her lab to teach him to reproduce the sounds of the English language. "I thought if he learned to communicate, I could ask him questions about how he sees the world."

B When Pepperberg began her dialogue with Alex, who died last September at the age of 31, many scientists believed animals were incapable of any thought. They were simply machines, robots programmed to react to stimuli but lacking the ability to think or feel. Any pet owner would disagree. We see the love in our dogs' eyes and know that, of course, they have thoughts and emotions. But such claims remain highly controversial. Gut instinct is not science, and it is all too easy to project human thoughts and feelings onto another creature. How, then, does a scientist prove that an animal is capable of thinking – that it is able to acquire information about the world and act on it? "That's why I started my studies with Alex," Pepperberg said. They were seated – she at her desk, he on top of his cage – in her lab, a windowless room about the size of a boxcar, at Brandeis University. Newspapers lined the

floor; baskets of bright toys were stacked on the shelves. They were clearly a team – and because of their work, the notion that animals can think is no longer so fanciful.

C Certain skills are considered key signs of higher mental abilities: good memory, a grasp of grammar and symbols, self-awareness, understanding others' motives, imitating others, and being creative. Bit by bit, in ingenious experiments, researchers have documented these talents in other species, gradually chipping away at what we thought made human beings distinctive while offering a glimpse of where our own abilities came from. Scrub jays know that other jays are thieves and that stashed food can spoil; sheep can recognize faces; chimpanzees use a variety of tools to probe termite mounds and even use weapons to hunt small mammals; dolphins can imitate human postures; the archerfish, which stuns insects with a sudden blast of water, can learn how to aim its squirt simply by watching an experienced fish perform the task. And Alex the parrot turned out to be a surprisingly good talker.

D Thirty years after the Alex studies began; Pepperberg and a changing collection of assistants were still giving him English lessons. The humans, along with two younger parrots, also served as Alex's flock, providing the social input all parrots crave. Like any flock, this one – as small as it was – had its share of drama. Alex dominated his fellow parrots, acted huffy at times around Pepperberg, tolerated the other female humans, and fell to pieces over a male assistant who dropped by for a visit. Pepperberg bought Alex in a Chicago pet store where she let the store's assistant pick him out because she didn't want other scientists saying later that she'd particularly chosen an especially smart bird for her work. Given that Alex's brain was the size of a shelled walnut, most researchers thought Pepperberg's interspecies communication study would be futile.

E "Some people actually called me crazy for trying this," she said. "Scientists thought that chimpanzees were better subjects, although, of course, chimps can't speak." Chimpanzees, bonobos, and gorillas have been taught to use sign language and symbols to communicate with us, often with impressive results. The bonobo Kanzi, for instance, carries his symbol-communication board with him so he can "talk" to his human researchers, and he has invented combinations of symbols to express his thoughts. Nevertheless, this is not the same thing as having an animal look up at you, open his mouth, and speak. Under Pepperberg's patient tutelage, Alex learned how to use his vocal tract to imitate almost one hundred English words, including the sounds for various foods, although he calls an apple a "banerry." "Apples taste a little bit like bananas to him, and they look a little bit like cherries, so Alex made up that word for them," Pepperberg said.

F It sounded a bit mad, the idea of a bird having lessons to practice, and willingly doing it. But after listening to and observing Alex, it was difficult to argue with Pepperberg's explanation for his behaviors. She wasn't handing him treats for the repetitious work or rapping him on the claws to make him say the sounds. "He has to hear the words over and over before he can correctly imitate them," Pepperberg said, after pronouncing "seven" for Alex a good dozen times in a row. "I'm not trying to see if Alex can learn a human language," she added. "That's never been the point. My plan always was to use his imitative skills to get a better understanding of avian cognition."

G In other words, because Alex was able to produce a close approximation of the sounds of some English words, Pepperberg could ask him questions about a bird's basic understanding of the world. She couldn't ask him what he was thinking about, but she could ask him about his knowledge of numbers, shapes, and colors. To demonstrate, Pepperberg carried Alex on her arm to a tall wooden perch in the middle of the room. She then retrieved a green key and a small green cup from a basket on a shelf. She held up the two items to Alex's eye. "What's the same?" she

asked. Without hesitation, Alex's beak opened: "Co-lor." "What's different?" Pepperberg asked. "Shape," Alex said. His voice had the digitized sound of a cartoon character.

Since parrots lack lips (another reason it was difficult for Alex to pronounce some sounds, such as ba), the words seemed to come from the air around him, as if a ventriloquist were speaking. But the words – and what can only be called the thoughts – were entirely his.

H For the next 20 minutes, Alex can through his tests, distinguish colors, shapes, sizes, and materials (wool versus wood versus metal). He did some simple arithmetic, such as accounting for the yellow toy blocks among a pile of mixed hues. And, then, as if to offer final proof of the mind inside his bird's brain, Alex spoke up. "Talk clearly!" he commanded when one of the younger birds Pepperberg was also teaching talked with wrong pronunciation. "Talk clearly!" "Don't be a smart aleck," Pepperberg said, shaking her head at him. "He knows all this, and he gets bored, so he interrupts the others, or he gives the wrong answer just to be obstinate. At this stage, he's like a teenager; he's moody, and I'm never sure what he'll do."

Questions 1-6

Do the following statements agree with the information given in Reading Passage 1?

In boxes 1-6 on your answer sheet, write

TRUE if the statement is true

FALSE if the statement is false

NOT GIVEN if the information is not given in the passage

1 Firstly, Alex has grasped quite a lot of vocabulary.

2 At the beginning of the study, Alex felt frightened in the presence of humans.

3 Previously, many scientists realized that the animal possesses the ability of thinking.

4 It has taken a long time before people get to know cognition existing in animals.

5 As Alex could approximately imitate the sounds of English words, he was capable of roughly answering Irene's questions regarding the world.

6 By breaking in other parrots as well as producing the incorrect answers, he tried to be focused.

Questions 7-10

Complete the following summary of the paragraphs of Reading Passage 1.

Using **NO MORE THAN THREE WORDS** from the Reading Passage 1 for each answer.

Write your answers in boxes 7-10 on your answer sheet.

After the training of Irene, Parrot Alex can use his vocal tract to pronounce more than **7**....., while other scientists believe that animals have no this advanced ability of thinking, they would rather teach **8**.....

Pepperberg clarified that she wanted to conduct a study concerning **9**..... but not to teach him to talk.

The store's assistant picked out a bird at random for her for the sake of avoiding other scientists saying that the bird is **10**..... afterward.

Questions 11-13

Answer the questions **11-13** below.

Choose **NO MORE THAN THREE WORDS AND/OR A NUMBER** from the passage for each answer.

11 What did Alex reply regarding the similarity of the subjects shown to him?

12 What is the problem of the young parrots except for Alex?

13 To some extent, through the way, he behaved what we can call him?

The Impact of the Potato

Jeff Chapman relates the story of history's most important vegetable.

The potato was first cultivated in South America between three and seven thousand years ago, though scientists believe they may have grown wild in the region as long as 13,000 years ago. The genetic patterns of potato distribution indicate that the potato probably originated in the mountainous west-central region of the continent.

Early Spanish chroniclers who misused the Indian word batata (sweet potato) as the name for the potato noted the importance of the tuber to the Incan Empire. The Incas had learned to preserve the potato for storage by dehydrating and mashing potatoes into a substance called Chuño. Chuño could be stored in a room for up to 10 years, providing excellent insurance against possible crop failures. As well as using the food as a staple crop, the Incas thought potatoes made childbirth easier and used it to treat injuries.

The Spanish conquistadors first encountered the potato when they arrived in Peru in 1532 in search of gold, and noted Inca miners eating Chuño. At the time the Spaniards failed to realise that the potato represented a far more important treasure than either silver or gold, but they did gradually begin to use potatoes as basic rations aboard their ships. After the arrival of the potato in Spain in 1570, a few Spanish farmers began to cultivate them on a small scale, mostly as food for livestock.

Throughout Europe, potatoes were regarded with suspicion, distaste and fear. Generally considered to be unfit for human consumption, they were used only as animal fodder and sustenance for the starving. In northern Europe, potatoes were primarily grown in botanical gardens as an exotic novelty. Even peasants refused to eat from a plant that produced ugly, misshapen tubers and that had come from a heathen civilization. Some felt that the potato plant's resemblance to plants in the nightshade family hinted that it was the creation of witches or devils.

In meat-loving England, farmers and urban workers regarded potatoes with extreme distaste. In 1662, the Royal Society recommended the cultivation of the tuber to the English government and the nation, but this recommendation had little impact. Potatoes did not become a staple until, during the food shortages associated with the Revolutionary Wars, the English government began to officially encourage potato cultivation. In 1795, the Board of Agriculture issued a pamphlet entitled "Hints Respecting the Culture and Use of Potatoes"; this was followed shortly by pro-potato editorials and potato recipes in *The Times*. Gradually, the lower classes began to follow the lead of the upper classes.

A similar pattern emerged across the English Channel in the Netherlands, Belgium and France. While the potato slowly gained ground in eastern France (where it was often the only crop remaining after marauding soldiers plundered wheat fields and vineyards), it did not achieve widespread acceptance until the late 1700s. The peasants remained suspicious, in spite of a 1771 paper from the Faculté de Paris testifying that the potato was not harmful but beneficial. The people began to overcome their distaste when the plant received the royal seal of approval: Louis XVI began to sport a potato flower in his buttonhole, and Marie-Antoinette wore the purple potato blossom in her hair.

Frederick the Great of Prussia saw the potato's potential to help feed his nation and lower the price of bread, but faced the challenge of overcoming the people's prejudice against the plant. When he issued a 1774 order for his subjects to grow potatoes as protection against famine, the town of Kolberg replied: "The things have neither smell nor taste, not even the dogs will eat them, so what use are they to us?" Trying a less direct approach to encourage his subjects to begin planting potatoes, Frederick used a bit of reverse psychology: he planted a royal field of potato

plants and stationed a heavy guard to protect this field from thieves. Nearby peasants naturally assumed that anything worth guarding was worth stealing, and so snuck into the field and snatched the plants for their home gardens. Of course, this was entirely in line with Frederick's wishes.

Historians debate whether the potato was primarily a cause or an effect of the huge population boom in industrial-era England and Wales. Prior to 1800, the English diet had consisted primarily of meat, supplemented by bread, butter and cheese. Few vegetables were consumed, most vegetables being regarded as nutritionally worthless and potentially harmful. This view began to change gradually in the late 1700s, The Industrial Revolution was drawing an ever-increasing percentage of the populace into crowded cities, where only the richest could afford homes with ovens or coal storage rooms, and people were working 12-16 hour days which left them with little time or energy to prepare food. High-yielding, easily prepared potato crops were the obvious solution to England's food problems.

Whereas most of their neighbours regarded the potato with suspicion and had to be persuaded to use it by the upper classes, the Irish peasantry embraced the tuber more passionately than anyone since the Incas. The potato was well suited to the Irish soil and climate, and its high yield suited the most important concern of most Irish farmers: to feed their families.

The most dramatic example of the potato's potential to alter population patterns occurred in Ireland, where the potato had become a staple by 1800. The Irish population doubled to eight million between 1780 and 1841, without any significant expansion of industry or reform of agricultural techniques beyond the widespread cultivation of the potato. Though Irish landholding practices were primitive in comparison with those of England, the potato's high yields allowed even the poorest farmers to produce more healthy food than they needed with scarcely any investment or hard labour. Even children could easily plant, harvest and cook potatoes, which of course required no threshing, curing or grinding. The abundance provided by potatoes greatly decreased infant mortality and encouraged early marriage.

Questions 1-5

Do the following statements agree with the information given in Reading Passage 1?

In boxes 1-5 on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

1 The early Spanish called potato as the Incan name 'Chuño'.

2 The purpose of Spanish coming to Peru was to find potatoes.

3 The Spanish believed that the potato had the same nutrients as other vegetables.

4 Peasants at that time did not like to eat potatoes because they were ugly,

5 The popularity of potatoes in the UK was due to food shortages during the war.

Questions 6-13

Complete the sentences below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 6-13 on your answer sheet.

6 In France, people started to overcome their disgust about potatoes because the King put a potato _____ in his buttonhole.

7 Frederick realised the potential of potatoes but he had to handle the _____ against potatoes from ordinary people.

- 8 The King of Prussia adopted some _____ psychology to make people accept potatoes.
- 9 Before 1800, the English people preferred eating _____ with bread, butter and cheese.
- 10 The obvious way to deal with England food problems was to grow high yielding potato _____.
- 11 The Irish _____ and climate suited potatoes well.
- 12 Between 1780 and 1841, based on the _____ of the potatoes, the Irish population doubled to eight million.
- 13 The potato's high yields helped the poorest farmers to produce much healthier food almost without _____ or hard physical work.

Ancient Storytelling

A It was told, we suppose, to people crouched around a fire: a tale of adventure, most likely relating some close encounter with death; a remarkable hunt, an escape from mortal danger; a vision, or something else out of the ordinary. Whatever its thread, the weaving of this story was done with a prime purpose. The listeners must be kept listening. They must not fall asleep. So, as the story went on, its audience should be sustained by one question above all. What happens next?

B The first fireside stories in human history can never be known. They were kept in the heads of those who told them. This method of storage is not necessarily inefficient. From documented oral traditions in Australia, the Balkans and other parts of the world we know that specialised storytellers and poets can recite from memory literally thousands of lines, in verse or prose, verbatim word for word. But while memory is rightly considered an art in itself, it is clear that the primary purpose of making symbols is to have a system of reminders or mnemonic cues – signs that assist us in recalling certain information in the mind's eye.

C In some Polynesian communities a notched memory stick may help to guide a storyteller through successive stages of recitation. But in other parts of the world, the activity of storytelling historically resulted in the development or even the invention of writing systems. One theory about the arrival of literacy in ancient Greece, for example, argues that the epic tales about the Trojan War and the wanderings of Odysseus – traditionally attributed to Homer – were just so enchanting to hear that they had to be preserved. So the Greeks, c.750-700BC, borrowed an alphabet from their neighbors in the eastern Mediterranean, the Phoenicians.

D The custom of recording stories on parchment and other materials can be traced in many manifestations around the world, from the priestly papyrus archives of ancient Egypt to the birch-bark scrolls on which the North American Ojibway Indians set down their creation-myth. It is a well-tryed and universal practice: so much so that to this day story time is probably most often associated with words on paper. The formal practice of narrating a story aloud would seem we assume to have given way to newspapers, novels and comic strips. This, however, is not the case. Statistically it is doubtful that the majority of humans currently rely upon the written word to get access to stories. So what is the alternative source?

E Each year, over 7 billion people will go to watch the latest offerings from Hollywood, Bollywood and beyond. The supreme storyteller of today is cinema. The movies, as distinct from still photography, seem to be an essentially modern phenomenon. This is an illusion, for there are, as we shall see, certain ways in which the medium of film is indebted to very old precedents of arranging 'sequences' of images. But any account of visual storytelling must be with the recognition that all storytelling beats with a deeply atavistic pulse: that is, a 'good story' relies upon formal patterns of plot and characterisation that have been embedded in the practice of storytelling over many generations.

F Thousands of scripts arrive every week at the offices of the major film studios. But aspiring screenwriters really need to look no further for essential advice than the fourth-century BC Greek Philosopher Aristotle. He left some incomplete lecture notes on the art of telling stories in various literary and dramatic modes, a slim volume known as *The Poetics*. Though he can never have envisaged the popcorn-fuelled actuality of a multiplex cinema, Aristotle is almost prescient about the key elements required to get the crowds flocking to such a cultural hub. He analyzed the process with cool rationalism. When a story enchants us, we lose the sense of where we are; we are drawn into the story so thoroughly that we forget it is a story being told. This is, in Aristotle's phrase, 'the suspension of disbelief.'

G We know the feeling. If ever we have stayed in our seats, stunned with grief, as the credits roll by, or for days after seeing that vivid evocation of horror have been nervous about taking a shower at home, then we have suspended disbelief. We have been caught, or captivated, in the storyteller's web. Did it all really happen? We really thought so for a while. Aristotle must have witnessed often enough this suspension of disbelief. He taught at Athens, the city where theater developed as a primary form of civic ritual and recreation. Two theatrical types of storytelling, tragedy and comedy, caused Athenian audiences to lose themselves in sadness and laughter respectively. Tragedy, for Aristotle, was particularly potent in its capacity to enlist and then purge the emotions of those watching the story unfold on the stage, so he tried to identify those factors in the storyteller's art that brought about such engagement. He had, as an obvious sample for analysis, not only the fifth-century BC masterpieces of Classical Greek tragedy written by Aeschylus, Sophocles and Euripides. Beyond them stood Homer, whose stories even then had canonical status: *The Iliad* and *The Odyssey* were already considered literary landmarks-stories by which all other stories should be measured. So what was the secret of Homer's narrative art?

H It was not hard to find. Homer created credible heroes. His heroes belonged to the past, they were mighty and magnificent, yet they were not, in the end, fantasy figures. He made his heroes sulk, bicker, cheat and cry. They were, in short, characters – protagonists of a story that an audience would care about, would want to follow, and would want to know what happens next. As Aristotle saw, the hero who shows a human side-some flaw or weakness to which mortals are prone is intrinsically dramatic by logging.

Questions 1-5

The Reading Passage has eight paragraphs A-H Which paragraph contains the following information?

Write the correct letter A-H, in boxes 1-5 on your answer sheet.

- 1** A misunderstanding of a modern way of telling stories
- 2** The typical forms mentioned for telling stories
- 3** The fundamental aim of storytelling
- 4** A description of reciting stories without any assistance
- 5** How to make story characters attractive

Questions 6-9

Write the correct letter, A, B or C in boxes 6-9 on your answer sheet.

Classify the following information as referring to

- A adopted the writing system from another country
 - B used organic materials to record stories
 - C used tools to help to tell stories
- 6** Egyptians
 - 7** Ojibway
 - 8** Polynesians
 - 9** Greek

Questions 10-13

Complete the sentences below with **ONE WORD ONLY** from the passage.

Write your answer in boxes 10-13 on your answer sheet.

10 Aristotle wrote a book on the art of storytelling called

11 Aristotle believed the most powerful type of story to move listeners is

12 Aristotle viewed Homers works as

13 Aristotle believed attractive heroes should have some

William Gilbert and Magnetism

A 16th and 17th centuries saw two great pioneers of modern science: Galileo and Gilbert. The impact of their findings is eminent. Gilbert was the first modern scientist, also the accredited father of the science of electricity and magnetism, an Englishman of learning and a physician at the court of Elizabeth. Prior to him, all that was known of electricity and magnetism was what the ancients knew, nothing more than that the: lodestone possessed magnetic properties and that amber and jet, when rubbed, would attract bits of paper or other substances of small specific gravity. However, he is less well-known than he deserves.

B Gilbert's birth predated Galileo. Born in an eminent local family in Colchester county in the UK, on May 24, 1544, he went to grammar school, and then studied medicine at St. John's College, Cambridge, graduating in 1573. Later he traveled in the continent and eventually settled down in London.

C He was a very successful and eminent doctor. All this culminated in his election to the president of the Royal Science Society. He was also appointed the personal physician to the Queen (Elizabeth I), and later knighted by the Queen. He faithfully served her until her death. However, he didn't outlive the Queen for long and died on December 10, 1603, only a few months after his appointment as personal physician to King James.

D Gilbert was first interested in chemistry but later changed his focus due to the large portion of the mysticism of alchemy involved (such as the transmutation of metal). He gradually developed his interest in physics after the great minds of the ancient, particularly about the knowledge the ancient Greeks had about lodestones, strange minerals with the power to attract iron. In the meantime, Britain became a major seafaring nation in 1588 when the Spanish Armada was defeated, opening the way to British settlement of America. British ships depended on the magnetic: compass, yet no one understood why it worked. Did the pole star attract it, as Columbus once speculated; or was there a magnetic mountain at the pole, as described in the Odyssey' which ships would never approach, because the sailors thought its pull would yank out all their iron nails and fitting? For nearly 20 years William Gilbert conducted ingenious experiments to understand magnetism. His works include On the Magnet and Magnetic Bodies, Great Magnet of the Earth.

E Gilbert's discovery was so important to modern physics. He investigated the nature of magnetism and electricity. He even coined the word "electric". Though the early beliefs of magnetism were also largely entangled with superstitions such as that rubbing garlic on lodestone can neutralize its magnetism, one example being that sailors even believed the smell of garlic would even interfere with the action of compass, which is why helmsmen were forbidden to eat it near a ship's compass. Gilbert also found that metals can be magnetized by rubbing materials such as fur, plastic or the like on them. He named the ends of a magnet "north pole" and "south pole". The magnetic poles can attract or repel, depending on polarity. In addition, however, ordinary iron is always attracted to a magnet. Though he started to study the relationship between magnetism and electricity, sadly he

didn't complete it. His research of static electricity using amber and jet only demonstrated that objects with electrical charges can work like magnets attracting small pieces of paper and stuff. It is a French guy named Du Fay that discovered that there are actually two electrical charges, positive and negative.

F He also questioned the traditional astronomical beliefs. Though a Copernican, he didn't express in his quintessential beliefs whether the earth is at the center of the universe or in orbit around the sun. However, he believed that stars are not equidistant from the earth, but have their own Earth-like planets orbiting around them. The earth is itself like a giant magnet, which is also why compasses always point north. They spin on an axis that is aligned with the earth's polarity. He even likened the polarity of the magnet to the polarity of the earth and built an entire magnetic philosophy on this analogy. In his explanation, magnetism was the soul of the earth. Thus a perfectly spherical lodestone, when aligned with the earth's poles, would wobble all by itself in 24 hours. Further, he also believed that suns and other stars wobble just like the earth does around a crystal core, and speculated that the moon might also be a magnet caused to orbit by its magnetic attraction to the earth. This was perhaps the first proposal that a force might cause a heavenly orbit.

G His research method was revolutionary in that he used experiments rather than pure logic and reasoning like the ancient Greek philosophers did. It was a new attitude toward scientific investigation. Until then, scientific experiments were not in fashion. It was because of this scientific attitude, together with his contribution to our knowledge of magnetism, that a unit of magneto-motive force, also known as magnetic potential, was named Gilbert in his honor. His approach of careful observation and experimentation rather than the authoritative opinion or deductive philosophy of others had laid the very foundation for modern science.

Questions 1-7

Reading passage 1 has seven paragraphs **A-G**.

Choose the correct heading for each paragraph from the list of headings below

Write the correct number i-x in boxes on your answer sheet.

List of Headings

- i Early years of Gilbert
- ii What was new about his scientific research method
- iii The development of chemistry
- iv Questioning traditional astronomy
- v Pioneers of the early science
- vi Professional and social recognition
- vii Becoming the president of the Royal Science Society
- viii The great works of Gilbert
- ix His discovery about magnetism
- x His change of focus

1 Paragraph A

2 Paragraph B

3 Paragraph C

4 Paragraph D

5 Paragraph E

6 Paragraph F

7 Paragraph G

Questions 8-10

Do the following statements agree with the information given in Reading Passage 1?

In boxes 8-10 on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

8 He is less famous than he should be.

9 He was famous as a doctor before he was employed by the Queen.

10 He lost faith in the medical theories of his time.

Questions 11-13

Choose **THREE** letters **A-F**.

Write your answers in boxes 11-13 on your answer sheet.

Which **THREE** of the following are parts of Gilbert's discovery?

- A** Metal can be transformed into another.
- B** Garlic can remove magnetism.
- C** Metals can be magnetized.
- D** Stars are at different distances from the earth.
- E** The earth wobbles on its axis.
- F** There are two charges of electricity

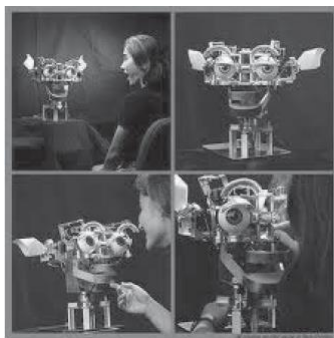
Man or Machine

A During July 2003, the Museum of Science in Cambridge, Massachusetts exhibited what Honda calls 'the world's most advanced humanoid robot', ASIMO (the Advanced Step in Innovative Mobility). Honda's brainchild is on tour in North America and delighting audiences wherever it goes. After 17 years in the making, ASIMO stands at four feet tall, weighs around 115 pounds and looks like a child in an astronaut's suit. Though it is difficult to see ASIMO's face at a distance, on closer inspection it has a smile and two large 'eyes' that conceal cameras. The robot cannot work autonomously — its actions are 'remote controlled' by scientists through the computer in its backpack. Yet watching ASIMO perform at a show in Massachusetts it seemed uncannily human. The audience cheered as ASIMO walked forwards and backward, side to side and up and downstairs. After the show, a number of people told me that they would like robots to play more of a role in daily life — one even said that the robot would be like 'another person'.



B While the Japanese have made huge strides in solving some of the engineering problems of human kinetics and bipedal movements, for the past 10 years scientists at MIT's former Artificial Intelligence (AI) lab (recently renamed the Computer Science and Artificial Intelligence Laboratory, CSAIL) have been making robots that can behave like humans and interact with humans. One of MIT's robots, Kismet, is an anthropomorphic head and has two eyes (complete with eyelids), ears, a mouth, and eyebrows. It has several facial expressions, including happy, sad, frightened and disgusted. Human interlocutors are able to read some of the robot's facial expressions, and often

change their behavior towards the machine as a result — for example, playing with it when it appears ‘sad’. Kismet is now in MIT’s museum, but the ideas developed here continue to be explored in new robots.



C Cog (short for Cognition) is another pioneering project from MIT’s former AI lab. Cog has a head, eyes, two arms, hands and a torso — and its proportions were originally measured from the body of a researcher in the lab. The work on Cog has been used to test theories of embodiment and developmental robotics, particularly getting a robot to develop intelligence by responding to its environment via sensors, and to learn through these types of interactions.



D MIT is getting furthest down the road to creating human-like and interactive robots. Some scientists argue that ASIMO is a great engineering feat but not an intelligent machine — because it is unable to interact autonomously with unpredictabilities in its environment in meaningful ways, and learn from experience. Robots like Cog and Kismet and new robots at MIT’s CSAIL and media lab, however, are beginning to do this.

E These are exciting developments. Creating a machine that can walk, make gestures and learn from its environment is an amazing achievement. And watch this space: these achievements are likely rapidly to be improved upon. Humanoid robots could have a plethora of uses in society, helping to free people from everyday tasks. In Japan, for example, there is an aim to create robots that can do the tasks similar to an average human, and also act in more sophisticated situations as firefighters, astronauts or medical assistants to the elderly in the workplace and in homes—partly in order to counterbalance the effects of an aging population.



F Such robots say much about the way in which we view humanity, and they bring out the best and worst of us. On one hand, these developments express human creativity — our ability to invent, experiment, and to extend our control over the world. On the other hand, the aim to create a robot like a human being is spurred on by dehumanized ideas — by the sense that human companionship can be substituted by machines; that humans lose their humanity when they interact with technology; or that we are little more than surface and ritual behaviors, that can be simulated with metal and electrical circuits.

Questions 1-6

Reading Passage 1 has six paragraphs, **A-F**.

Which paragraph contains the following information?

Write the correct letter, **A-F**, in boxes 1-6 on your answer sheet.

NB you may use any letter more than once

- 1 different ways of using robots
- 2 a robot whose body has the same proportion as that of an adult
- 3 the fact that humans can be copied and replaced by robots
- 4 a comparison between ASIMO from Honda and other robots
- 5 the pros and cons of creating robots
- 6 a robot that has eyebrows

Questions 7-13

Complete the following summary of the paragraphs of Reading Passage 1

Using **NO MORE THAN TWO WORDS** from the passage for each answer.

Write your answers in boxes **7-13** on your answer sheet.

In 2003, Massachusetts displayed a robot named ASIMO which was invented by Honda, after a period of **7** _____ in the making. The operating information is stored in the computer in its **8** _____ so that scientists can control ASIMO's movement. While Japan is making great progress, MIT is developing robots that are human-like and can **9** _____ humans. What is special about Kismet is that it has different **10** _____ which can be read by human interlocutors. **11** _____ is another robot from MIT, whose body's proportion is the same as an adult. By responding to the surroundings through **12** _____, it could develop its **13** _____.

The Innovation of Grocery Stores

A At the beginning of the 20th century, grocery stores in the United States were full-service. A customer would ask a clerk behind the counter for specific items and the clerk would package the items, which were limited to dry goods. If they want to save some time, they have to ask a delivery boy or by themselves to send the note of what they want to buy to the grocery store first and then go to pay for the goods later. These grocery stores usually carried only one brand of each good. There were early chain stores, such as the A&P Stores, but these were all entirely full-service and very time-consuming.

B In 1885, a Virginia boy named Clarence Saunders began working part-time as a clerk in a grocery store when he was 14 years old, and quit school when the shopkeeper offered him full-time work with room and board. Later he worked in an Alabama coke plant and in a Tennessee sawmill before he returned to the grocery business. By 1900, when he was nineteen years old, he was earning \$30 a month as a salesman for a wholesale grocer. During his years working in the grocery stores, he found that it was very inconvenient and inefficient for people to buy things because more than a century ago, long before there were computers, shopping was done quite differently than it is today. Entering a store, the customer would approach the counter (or wait for a clerk to become available) and place an order, either verbally or, as was often the case for boys running errands, in the form of a note or list. While the

customer waited, the clerk would move behind the counter and throughout the store, select the items on the list—some from shelves so high that a long-handled grasping device had to be used—and bring them back to the counter to be tallied and bagged or boxed. The process might be expedited by the customer calling or sending in the order beforehand, or by the order being handed by a delivery boy on a bike, but otherwise it did not vary greatly. Saunders, a flamboyant and innovative man, noticed that this method resulted in wasted time and expense, so he came up with an unheard-of solution that would revolutionize the entire grocery industry: he developed a way for shoppers to serve themselves.

C So in 1902 he moved to Memphis where he developed his concept to form a grocery wholesale cooperative and a full-service grocery store. For his new 'cafeteria grocery', Saunders divided his grocery into three distinct areas: 1) A front 'lobby' forming an entrance and exit and checkouts at the front. 2) A sales department, which was specially designed to allow customers to roam the aisles and select their own groceries. Removing unnecessary clerks, creating elaborate aisle displays, and rearranging the store to force customers to view all of the merchandise and over the shelving and cabinets units of sales department were 'galleries' where supervisors were allowed to keep an eye on the customers while not disturbing them. 3) And another section of his store is the room only allowed for the clerks which was called the 'stockroom' or 'storage room' where large refrigerators were situated to keep fresh products from being perishable. The new format allowed multiple customers to shop at the same time, and led to the previously unknown phenomenon of impulse shopping. Though this format of grocery market was drastically different from its competitors, the style became the standard for the modern grocery store and later supermarkets.

D On September 6, 1916, Saunders launched the self-service revolution in the USA by opening the first self-service Piggly Wiggly store, at 79 Jefferson Street in Memphis, Tennessee, with its characteristic turnstile at the entrance. Customers paid cash and selected their own goods from the shelves. It was unlike any other grocery store of that time. Inside a Piggly Wiggly, shoppers were not at the mercy of shop clerks. They were free to roam the store, check out the merchandise and get what they needed with their own two hands and feet. Prices on items at Piggly Wiggly were clearly marked. No one pressured customers to buy milk or pickles. And the biggest benefit at the Piggly Wiggly was that shoppers saved money. Self-service was a positive all around. 'It's good for both the consumer and retailer because it cuts costs,' noted George T. Haley, a professor at the University of New Haven and director of the Center for International Industry Competitiveness. 'If you looked at the way grocery stores were run previous to Piggly Wiggly and Alpha Beta, what you find is that there was a tremendous amount of labor involved, and labor is a major expense.' Piggly Wiggly cut the fat.

E Piggly Wiggly and the self-service concept took off. Saunders opened nine stores in the Memphis area within the first year of business. Consumers embraced the efficiency, the simplicity and most of all the lower food prices. Saunders soon patented his self-service concept, and began franchising Piggly Wiggly stores. Thanks to the benefits of self-service and franchising, Piggly Wiggly ballooned to nearly 1, 300 stores by 1923. Piggly Wiggly sold \$100 million—worth \$1. 3 billion today—in groceries, making it the third-biggest grocery retailer in the nation. The company's stock was even listed on the New York Stock Exchange, doubling from late 1922 to March 1923. Saunders had his hands all over Piggly Wiggly. He was instrumental in the design and layout of his stores. He even invented the turnstile.

F However Saunders was forced into bankruptcy in 1923 after a dramatic spat with the New York Stock Exchange and he went on to create the "Clarence Saunders sole-owner-of-my-name" chain, which went into bankruptcy.

G Until the time of his death in October 1953, Saunders was developing plans for another automatic store system called the Foodelectric. But the store, which was to be located two blocks from the first Piggly Wiggly store, never opened. But his name was well-remembered along with the name Piggly Wiggly.

Questions 1-5

The Reading Passage has seven paragraphs **A-G**.

Which paragraph contains the following information?

Write the correct letter **A-G**, in boxes **1-5** on your answer sheet.

NB You may use any letter more than once.

- 1** How Clarence Saunders' new idea had been carried out.
- 2** Introducing the modes and patterns of groceries before his age.
- 3** Clarence Saunders declared bankruptcy a few years later.
- 4** Descriptions of Clarence Saunders' new conception.
- 5** The booming development of his business.

Questions 6-10

Answer the questions below.

Write **ONLY ONE WORD AND/OR A NUMBER** from the passage for each answer.

- 6** When Clarence Saunders was an adolescent, he took a job as a _____ in a grocery store.
- 7** In the new innovation of grocery store, most of the clerks' work before was done by _____.
- 8** In Saunders' new grocery store, the section where customers finish the payment was called _____.
- 9** Another area in his store which behind the public area was called the _____, where only internal staff could access.
- 10** At _____ where customers were under surveillance.

Questions 11-13

Choose the correct letter, **A, B, C** or **D**.

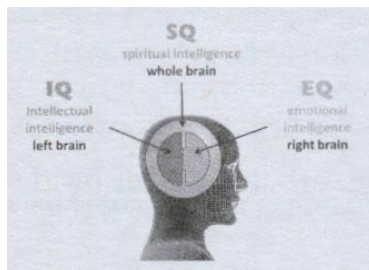
Write your answers in boxes **11-13** on your answer sheet.

- 11** Why did Clarence Saunders want to propel the innovation of grocery stores at his age?
 - A** Because he was an enthusiastic and creative man.
 - B** Because his boss wanted to reform the grocery industry.
 - C** Because he wanted to develop its efficiency and make great profit as well.
 - D** Because he worried about the future competition from the industry.
- 12** What happened to Clarence Saunders' first store of Piggly Wiggly?
 - A** Customers complained about its impracticality and inconvenience.
 - B** It enjoyed a great business and was updated in the first twelve months.
 - C** It expanded to more than a thousand franchised stores during the first year.
 - D** Saunders were required to have his new idea patented and open more stores.
- 13** What left to Clarence Saunders after his death in 1953?
 - A** A fully automatic store system opened soon near his first store.
 - B** The name of his store the Piggly Wiggly was very popular at that time.
 - C** His name was usually connected with his famous shop the Piggly Wiggly in the following several years.
 - D** His name was painted together with the name of his famous store.

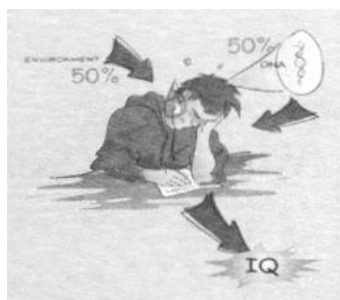
The Research for Intelligence

A In Robert Plomin's line of work, patience is essential. Plomin, a behavioral geneticist at the Institute of Psychiatry in London, wants to understand the nature of intelligence. As part of his research, he has been watching

thousands of children grow up. Plomin asks the children questions such as ‘What do water and milk have in common?’ and ‘In what direction does the sun set?’ At first he and his colleagues quizzed the children in person or over the telephone. Today many of those children are in their early teens, and they take their tests on the Internet. In one sense, the research has been a rousing success. The children who take the tests are all twins, and throughout the study identical twins have tended to get scores closer to each other than those of non-identical twins, who in turn have closer scores than unrelated children. These results—along with similar ones from other studies—make clear to the scientists that genes have an important influence on how children score on intelligence tests.



B But Plomin wants to know more. He wants to find the specific genes that are doing the influencing. And now he has a tool for pinpointing genes that he could not have even dreamed of when he began quizzing children. Plomin and his colleagues have been scanning the genes of his subjects with a device called a micro-array, a small chip that can recognize half a million distinctive snippets of DNA. The combination of this powerful tool with a huge number of children to study meant that he could detect genes that had only a tiny effect on the variation in scores.



C Still, when Plomin and his co-workers unveiled the results of their micro-array study—the biggest dragnet for intelligence-linked genes ever undertaken—they were underwhelming. The researchers found only six genetic markers that showed any sign of having an influence on the test scores. When they ran stringent statistical tests to see if the results were flukes, only one gene passed. It accounted for 0.4 percent of variation in the scores. And to cap it all off, no one knows what the gene does in the body. ‘It’s a real drag in some ways,’ Plomin says.

D Plomin’s experience is a typical one for scientists who study intelligence. Along with using micro-arrays, they are employing brain scans and other sophisticated technologies to document some of the intricate dance steps that genes and environment take together in the development of intelligence. They are beginning to see how differences in intelligence are reflected in the structure and function of the brain. Some scientists have even begun to build a new vision of intelligence as a reflection of the ways in which information flows through the brain. But for all these advances, intelligence remains a profound mystery. ‘It’s amazing the extent to which we know very little,’ says Wendy Johnson, a psychologist at the University of Minnesota.

E In some ways, intelligence is very simple. ‘It’s something that everybody observes in others,’ says Eric Turkheimer of the University of Virginia. Everybody knows that some people are smarter than others, whatever it means technically. It’s something you sense in people when you talk to them. Yet that kind of gut instinct does not

translate easily into a scientific definition. In 1996 the American Psychological Association issued a report on intelligence, which stated only that ‘individuals differ from one another in their ability to understand complex ideas, to adapt effectively to the environment, to learn from experience, to engage in various forms of reasoning, to overcome obstacles by taking thought.’



F To measure these differences, psychologists in the early 1900s invented tests of various kinds of thought, such as math, spatial reasoning and verbal skills. To compare scores on one type of test to those on another, some psychologists developed standard scales of intelligence. The most familiar of them is the intelligence quotient, which is produced by setting the average score at 100. IQ scores are not arbitrary numbers, however. Psychologists can use them to make strong predictions about other features of people’s lives. It is possible to make reasonably good predictions, based on IQ scores in childhood, about how well people will tire in school and in the workplace. People with high IQs even tend to live longer than average. ‘If you have an IQ score, does that tell you everything about a person’s cognitive strengths and weaknesses? No,’ says Richard J. Haier of the University of California, Irvine. But even a simple number has the potential to say a lot about a person. ‘When you go see your doctor, what’s the first thing that happens? Somebody takes your blood pressure and temperature. So you get two numbers. No one would say blood pressure and temperature summarize everything about your health, but they are key numbers.’



G Then what underlies an intelligence score? ‘It’s certainly tapping something,’ says Philip Shaw, a psychiatrist at the National Institute of Mental Health (NIMH). The most influential theory of what the score reflects is more than a century old. In 1904 psychologist Charles Spearman observed that people who did well on one kind of test tended to do well on others. The link from one score to another was not very tight, but Spearman saw enough of a connection to declare that it was the result of something he called a g-factor, short for general intelligence factor. How general intelligence arose from the brain, Spearman could not say. In recent decades, scientists have searched for an answer by finding patterns in the test scores of large groups of people. Roughly speaking, there are two possible sources for these variations. Environmental influences—anything from the way children are raised by their parents to the diseases they may suffer as they develop—are one source. Genes are another. Genes may shape the brain in ways that make individuals better or worse at answering questions on intelligence tests.

Questions 1-6

The reading passage has seven paragraphs, **A-G**

Choose the correct heading for paragraphs **B-G** from the list below.

Write the correct number, **i-x**, in boxes **1-6** on your answer sheet.

List of Headings

- i Low probability triggers unpersuasive findings

- ii Understanding of intelligence remains limited
- iii Difficulty in accurately defining intelligence
- iv People with high IQ seldom fall sick
- v An innovative appliance to improve the probe
- vi The financial cost of a new research
- vii Why an indicator is imperfect but referable
- viii Genes mean extra when compared with environment
- ix A vital indicator for kids' intelligence performance
- x Multiple factors involved in intelligence

Example

Answer

Paragraph A

ix

1 Paragraph B

2 Paragraph C

3 Paragraph D

4 Paragraph E

5 Paragraph F

6 Paragraph G

Questions 7-10

Use the information in the passage to match the people (listed **A-F**) with opinions or deeds below. Write the appropriate letters **A-F** in boxes **7-10** on your answer sheet.

A Plomin

B Philip Shawn

C Eric Turkheimer

D Charles Spearman

E Richard J. Haier

F Wendy Johnson

7 A full conclusion can be hardly reached just by the one example in IQ test.

8 It is not easy to exclude the occasionality existed in the research.

9 Humans still have more to explore in terms of the real nature of intelligence.

10 It is quite difficult to find the rear origins where the general intelligence comes.

Questions 11-13

Summary

Complete the following summary of the paragraphs of Reading Passage 1, using **NO MORE THAN THREE WORDS** from the passage for each answer.

Write your answers in boxes 11-13 on your answer sheet.

Many researchers including Plomin have faced with the typical challenge when **11** _____ are implemented. They try to use all possible methods to record certain **12** _____ performed both by genes and environment which contributes to the progress of intelligence. The relationship between intelligence and brain become their targeted area. What's more, according to some researchers, intelligence is regarded to be **13** _____ of how messages are transmitted in the brain.

Biomimetic Design

What has fins like a whale, skin like a lizard, and eyes like a moth? The future of engineering. Andrew Parker, an evolutionary biologist, knelt in the baking red sand of the Australian outback just south of Alice Springs and eased

the right hind leg of a thorny devil into a dish of water.

A 'Its back is completely drenched!' Sure enough, after 30 seconds, water from the dish had wicked up the lizard's leg and was glistening all over its prickly hide. In a few seconds more water reached its mouth, and the lizard began to smack its jaws with evident satisfaction. It was, in essence, drinking through its foot. Given more time, the thorny devil can perform this same conjuring trick on a patch of damp sand—a vital competitive advantage in the desert. Parker had come here to discover precisely how it does this, not from purely biological interest, but with a concrete purpose in mind: to make a thorny-devil-inspired device that will help people collect lifesaving water in the desert. 'The water's spreading out incredibly fast!' he said, as drops from his eyedropper fell onto the lizard's back and vanished, like magic. 'Its skin is far more hydrophobic than I thought. There may well be hidden capillaries, channeling the water into the mouth.'

B Parker's work is only a small part of an increasingly vigorous, global biomimetics movement. Engineers in Bath, England, and West Chester, Pennsylvania, are pondering the bumps on the leading edges of humpback whale flukes to learn how to make airplane wings for more agile flight. In Berlin, Germany, the fingerlike primary feathers of raptors are inspiring engineers to develop wings that change shape aloft to reduce drag and increase fuel efficiency. Architects in Zimbabwe are studying how termites regulate temperature, humidity, and airflow in their mounds in order to build more comfortable buildings, while Japanese medical researchers are reducing the pain of an injection by using hypodermic needles edged with tiny serrations, like those on a mosquito's proboscis, minimizing nerve stimulation.

C Ronald Fearing, a professor of electrical engineering at the University of California, Berkeley, has taken on one of the biggest challenges of all: to create a miniature robotic fly that is swift, small, and maneuverable enough for use in surveillance or search-and-rescue operations. Fearing made his own, one of which he held up with tweezers for me to see, a gossamer wand some 11 millimeters long and not much thicker than a cat's whisker. Fearing has been forced to manufacture many of the other minute components of his fly in the same way, using a micromachining laser and a rapid prototyping system that allows him to design his minuscule parts in a computer, automatically cut and cure them overnight, and assemble them by hand the next day under a microscope.

D With the microlaser he cuts the fly's wings out of a two-micron polyester sheet so delicate that it crumples if you breathe on it and must be reinforced with carbon-fiber spars. The wings on his current model flap at 275 times per second—faster than the insect's own wings—and make the blowfly's 'signature buzz'. 'Carbon fiber outperforms fly chitin,' he said, with a trace of self-satisfaction. He pointed out a protective plastic box on the lab bench, which contained the fly-bot itself, a delicate, origami-like framework of black carbon-fiber struts and hairlike wires that, not surprisingly, looked nothing like a real fly. A month later it achieved liftoff in a controlled flight on a boom. Fearing expects the fly-bot to hover in two or three years, and eventually to bank and dive with fly-like virtuosity.

E Stanford University roboticist Mark Cutkosky designed a gecko-inspired climber that he christened Stickybot. In reality, gecko feet aren't sticky—they're dry and smooth to the touch—and owe their remarkable adhesion to some two billion spatula-tipped filaments per square centimeter on their toe pads, each filament only a hundred nanometers thick. These filaments are so small, in fact, that they interact at the molecular level with the surface on which the gecko walks, tapping into the low-level van der Waals forces generated by molecules' fleeting positive and negative charges, which pull any two adjacent objects together. To make the toe pads for Stickybot, Cutkosky

and doctoral student Sangbae Kim, the robot's lead designer, produced a urethane fabric with tiny bristles that end in 30-micrometer points. Though not as flexible or adherent as the gecko itself, they hold the 500-gram robot on a vertical surface.

F Cutkosky endowed his robot with seven-segmented toes that drag and release just like the lizard's, and a gecko-like stride that snugs it to the wall. He also crafted Stickybot's legs and feet with a process he calls shape deposition manufacturing (SDM), which combines a range of metals, polymers, and fabrics to create the same smooth gradation from stiff to flexible that is present in the lizard's limbs and absent in most man-made materials. SDM also allows him to embed actuators, sensors, and other specialized structures that make Stickybot climb better. Then he noticed in a paper on gecko anatomy that the lizard had branching tendons to distribute its weight evenly across the entire surface of its toes. Eureka. 'When I saw that, I thought, Wow, that's great!' He subsequently embedded a branching polyester cloth 'tendon' in his robot's limbs to distribute its load in the same way.

G Stickybot now walks up vertical surfaces of glass, plastic, and glazed ceramic tile, though it will be some time before it can keep up with a gecko. For the moment it can walk only on smooth surfaces, at a mere four centimeters per second, a fraction of the speed of its biological role model. The dry adhesive on Stickybot's toes isn't self-cleaning like the lizard's either, so it rapidly clogs with dirt. 'There are a lot of things about the gecko that we simply had to ignore,' Cutkosky says. Still, a number of real-world applications are in the offing. The Department of Defense's Defense Advanced Research Projects Agency (DARPA), which funds the project, has it in mind for surveillance: an automaton that could slink up a building and perch there for hours or days, monitoring the terrain below. Cutkosky hypothesizes a range of civilian uses. 'I'm trying to get robots to go places where they've never gone before,' he told me. 'I would like to see Stickybot have a real-world function, whether it's a toy or another application. Sure, it would be great if it eventually has a lifesaving or humanitarian role. . . '

H For all the power of the biomimetics paradigm, and the brilliant people who practice it, bio-inspiration has led to surprisingly few mass-produced products and arguably only one household word—Velcro, which was invented in 1948 by Swiss chemist George de Mestral, by copying the way cockleburs clung to his dog's coat. In addition to Cutkosky's lab, five other high-powered research teams are currently trying to mimic gecko adhesion, and so far none has come close to matching the lizard's strong, directional, self-cleaning grip. Likewise, scientists have yet to meaningfully re-create the abalone nanostructure that accounts for the strength of its shell, and several well-funded biotech companies have gone bankrupt trying to make artificial spider silk.

Questions 1-7

Do the following statements agree with the information given in Reading Passage 1?

In boxes **1-7** on your answer sheet, write

TRUE if the statement is true

FALSE if the statement is false

NOT GIVEN if the information is not given in the passage

- 1** Andrew Parker failed to make effective water device which can be used in desert.
- 2** Skin of lizard is easy to get wet when it contacts water.
- 3** Scientists apply inspiration from nature into many artificial engineering.
- 4** Tiny and thin hair under gecko's feet allows it to stick to the surface of object.
- 5** When gecko climbs downward, its feet release a certain kind of chemical to make them adhesive.
- 6** Famous cases stimulate a large number of successful products of biomimetics in real life.
- 7** Velcro is well-known for its bionics design.

Questions 8-10

Filling the blanks below.

Write **NO MORE THAN THREE WORDS AND/OR A NUMBER** from the passage for each question of robot below.

Ronald Fearing was required to fabricate tiny components for his robotic fly in **8**____ by specialized techniques. The robotic fly's main structure outside is made of **9**____ and long and thin wires which make it unlike fly at all. Cutkosky applied an artificial material in stickybot's **10**____ as a tendon to split pressure like lizard's does.

Questions 11-13

Fill the blanks below.

Write **NO MORE THAN THREE WORDS AND/OR A NUMBER** from the passage for each answer about facts of sticky bit.

- 11** Stickybot's feet doesn't have _____ function which makes it only be able to walk on smooth surfaces.
- 12** DARPA are planning to use stickybot for _____ .
- 13** Cutkosky assume that stickybot finally has potential in _____ or other human-related activities.

Passage 2

Novice and Expert-Becoming an Expert

Expertise is commitment coupled with creativity. Specifically, it is the commitment of time, energy, and resources to a relatively narrow field of study and the creative energy necessary to generate new knowledge in that field. It takes a considerable amount of time and regular exposure to a large number of cases to become an expert.

A An individual enters a field a study as a novice. The novice needs to learn the guiding principles and rules of a given task in order to perform that task. Concurrently, the novice needs to be exposed to specific cases, or instances, that test the boundaries of such heuristics. Generally, a novice will find a mentor to guide her through the process. A fairly simple example would be someone learning to play chess. The novice chess player seeks a mentor to teach her the object of the game, the number of spaces, the names of the pieces, the function of each piece, how each piece is moved, and the necessary conditions for winning or losing the game.

B In time, and with much practice, the novice begins to recognize patterns of behavior within cases and, thus, becomes a journeyman. With more practice and exposure to increasingly complex cases, the journeyman finds patterns not only with cases but also between cases. More importantly, the journeyman learns that these patterns often repeat themselves over time. The journeyman still maintains regular contact with a mentor to solve specific problems and learn more complex strategies. Returning to the example of the chess player, the individual begins to learn patterns of opening moves, offensive and defensive game-playing strategies, and patterns of victory and defeat.

C When a journeyman starts to make and test hypotheses about future behavior based on past experiences, she begins the next transition. Once she creatively generates knowledge, rather than simply matching superficial patterns, she becomes an expert. At this point, she is confident in her knowledge and no longer needs a mentor as a guide – she becomes responsible for her own knowledge. In the chess example, once a journeyman begins competing against experts, makes predictions based on patterns, and tests those predictions against actual behavior, she is generating new knowledge and a deeper understanding of the game. She is creating her own cases rather than relying on the cases of others.

The Power of Expertise

D An expert perceives meaningful patterns in her domain better than non-experts. Where a novice perceives random or disconnected data points, an expert connects regular patterns within and between cases. This ability to identify patterns is not an innate perceptual skill; rather it reflects the organization of knowledge after exposure to and experience with thousands of cases. Experts have a deeper understanding of their domains than novices do, and utilize higher-order principles to solve problems. A novice, for example, might group objects together by color or size, whereas an expert would group the same objects according to their function or utility. Experts comprehend the meaning of data and weigh variables with different criteria within their domains better than novices. Experts recognize variables that have the largest influence on a particular problem and focus their attention on those variables.

E Experts have better domain-specific short-term and long-term memory than novices do. Moreover, experts perform tasks in their domains faster than novices and commit fewer errors while problem solving. Interestingly, experts go about solving problems differently than novices. Experts spend more time thinking about a problem to fully understand it at the beginning of a task than do novice, who immediately seek to find a solution. Experts use

their knowledge of previous cases as a context for creating mental models to solve given problems.

F Better at self-monitoring than novices, experts are more aware of instances where they have committed errors or failed to understand a problem. Experts check their solutions more often than novices and recognize when they are missing information necessary for solving a problem. Experts are aware of the limits of their domain knowledge and apply their domain's heuristics to solve problems that fall outside of their experience base.

The Paradox of Expertise

G The strengths of expertise can also be weakness. Although one would expect experts to be good forecasters, they are not particularly good at making predictions about the future. Since the 1930s, researchers have been testing the ability of experts to make forecasts. The performance of experts has been tested against actuarial tables to determine if they are better at making predictions than simple statistical models. Seventy years later, with more than two hundred experiments in different domains, it is clear that the answer is no. If supplied with an equal amount of data about a particular case, an actuarial table is as good, or better, than an expert at making calls about the future. Even if an expert is given more specific case information than is available to the statistical model, the expert does not tend to outperform the actuarial table.

H Theorists and researchers differ when trying to explain why experts are less accurate forecasters than statistical models. Some have argued that experts, like all humans, are inconsistent when using mental models to make predictions. A number of researchers point to human biases to explain unreliable expert predictions. A number of researchers point to human biases to explain unreliable expert predictions. During the last 30 years, researchers have categorized, experimented, and theorized about the cognitive aspects of forecasting. Despite such efforts, the literature shows little consensus regarding the causes or manifestation of human bias.

Questions 14-18

Complete the flow chart.

Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

Write your answers in boxes **14-18** on your answer sheet.

From a novice to an expert

Novice:

needs **14** and to perform a given task;

exposed to specific cases;

guided by a **15** Through learning

Journeyman:

starts to identify **16** within and between cases;

often exposed to **17** cases;

contacts a mentor when facing difficult problems

Expert:

Create predictions and new **18**

performs task independently without the help of a mentor

Questions 19-23

Do the following statements agree with the information given in Reading Passage 2?

In boxes **19-23** on your answer sheet, write

TRUE if the statement is true

FALSE if the statement is false

NOT GIVEN if the information is not given in the passage

19 Novices and experts use the same system to classify objects.

20 A novice's training is focused on memory skills.

21 Experts have higher efficiency than novices when solving problems in their own field.

22 When facing a problem, a novice always tries to solve it straight away.

23 Experts are better at recognizing their own mistakes and limits

Questions 24-26

Complete the following summary of the paragraphs of Reading Passage 2.

Using **NO MORE THAN TWO WORDS** from the Reading Passage for each answer.

Write your answers in boxes **24-26** on your answer sheet.

Though experts are quite effective at solving problems in their own domains, their strengths can also be turned against them. Studies have shown that experts are less **24** at making predictions than statistical models. Some researchers theorise it is because experts can also be inconsistent like all others. Yet some believe it is due to **25**....., but there isn't a great deal of **26**..... as to its cause and manifestation.

Movie of Metropolis

...being the science-fiction film that is steadily becoming a fact

A When German director Fritz Lang visited the United States in 1924, his first glimpse of the country was a night-time view of the New York skyline from the deck of an ocean liner. This, he later recalled, was the direct inspiration for what is still probably the most innovative and influential science-fiction film ever made – Metropolis.

B Metropolis is a bleak vision of the early twenty-first century that is at once both chilling and exhilarating. This spectacular city of the future is a technological marvel of high-rise buildings connected by elevated railways and airships. It's also a world of extreme inequality and social division. The workers live below ground and exist as machines working in an endless routine of mind-numbing 10-hour shifts while the city's elite lead lives of luxury high above. Presiding over them all is the Master of Metropolis, John Fredersen, whose sole satisfaction seems to lie in the exercise of power.

C Lang's graphic depiction of the future is conceived in almost totally abstract terms. The function of the individual machines is never defined. Instead, this mass of dials, levers and gauges symbolically stands for all machines and all industry, with the workers as slave-like extensions of the equipment they have to operate. Lang emphasizes this idea in the famous shift-change sequence at the start of the movie when the workers walk in zombie-like geometric ranks, all dressed in the same dark overalls and all exhibiting the same bowed head and dead-eyed stare. An extraordinary fantasy sequence sees one machine transformed into a huge open-jawed statue which then literally swallows them up.

D On one level the machines and the exploited workers simply provide the wealth and services which allow the elite to live their lives of leisure, but on a more profound level, the purpose of all this demented industry is to serve itself. Power, control and the continuance of the system from one 10-hour shift to the next is all that counts. The city consumes people and their labour and in the process becomes a perverse parody of a living being.

E It is enlightening, I think, to relate the film to the modern global economy in which multinational corporations now routinely close their factories in one continent so that they can take advantage of cheap labour in another. Like the industry in Metropolis, these corporations' goals of increased efficiency and profits have little to do with the

welfare of the majority of their employees or that of the population at large. Instead, their aims are to sustain the momentum of their own growth and to increase the monetary rewards to a tiny elite – their executives and shareholders. Fredersen himself is the essence of the big company boss: Rupert Murdoch would probably feel perfectly at home in his huge skyscraper office with its panoramic view of the city below. And it is important that there is never any mention of government in Metropolis – the whole concept is by implication obsolete. The only people who have power are the supreme industrialist, Fredersen, and his magician/scientist cohort Rotwang.

F So far so good: when the images are allowed to speak for themselves the film is impeccable both in its symbolism and in its cynicism. The problem with Metropolis is its sentimental storyline, which sees Freder, Fredersen's son, instantly falling in love with the visionary Maria. Maria leads an underground pseudo-religious movement and preaches that the workers should not rebel but should await the arrival of a 'Mediator' between the 'Head' (capital) and the 'Hands' (labour). That mediator is the 'Heart' – love, as embodied, finally, by Freder's love of Maria and his father's love of him.

G Lang wrote the screenplay in collaboration with his then-wife Thea von Harbou. In 1933 he fled from the Nazis (and continued a very successful career in Hollywood). She stayed in Germany and continued to make films under the Hitler regime. There is a constant tension within the film between the too-tidy platitudes of von Harbou's script and the uncompromisingly caustic vigour of Lang's imagery.

H To my mind, both in Metropolis and in the real world, it's not so much that the 'Head' and 'Hands' require a 'Heart' to mediate between them but that the 'Hands' need to develop their own 'Head', their own political consciousness, and act accordingly – through the ballot box, through buying power and through a skeptical resistance to the materialistic fantasies of the Fredersens.

I All the same, Metropolis is probably more accurate now as a representation of industrial and social relations than it has been at any time since its original release. And Fredersen is certainly still the most potent movie symbol of the handful of elusive corporate figureheads who increasingly treat the world as a Metropolis-like global village.

Questions 14-26

Do the following statements agree with the claims of the writer in Reading Passage 2?

In boxes **14-17** on your answer sheet, write

YES if the statement is true

NO if the statement is false

NOT GIVEN if the information is not given in the passage

14 The inspiration of the movie Metropolis comes from the director's visit in the USA in 1924.

15 The Master of Metropolis, John Fredersen, is portrayed from an industrialist that the director met in the US.

16 The start of the movie exhibits the workers working in full energy.

17 The director and his wife got divorced because his wife decided to stay in Germany.

Questions 17-22

Complete the summary below.

Using **NO MORE THAN TWO WORDS** from the Reading Passage for each answer.

Write your answers in boxes **31-36** on your answer sheet.

The director depicts a world of inequality and **17**.....In the future, the mindless masses of workers living underground are treated as **18**.....And the master of them is **19**....., who is in charge of the whole city. The writer claims that the director, Fritz Lang, presents the movie in an **20**.....term, where the **21**.....of

the individual machines is not defined. Besides the writer compares the film to the modern global economy in which multinational corporations concern more about the growing **22**.....and money.

Questions 23-26

Choose the correct letter, **A, B, C or D**.

Write your answers in boxes **23-26** on your answer sheet.

23 The first sentence in paragraph B indicates

- A the author's fear about technology
- B the inspiration of the director
- C the contradictory feelings toward future
- D the city elite's well management of the workers

24 Why the function of the individual machines is not defined?

- A Because Lang sticks to theme in a symbolic way.
- B Because workers are more important to exploit.
- C Because the fantasy sequence is difficult to take.
- D Because the focus of the movie is not about machines.

25 The writer's purpose in paragraph five is to

- A emphasize the multinational corporations' profit-oriented goal.
- B compare the movie with the reality in the modern global economy
- C exploit the difference between fantasy and reality
- D enlighten the undeveloped industry

26 What is the writer's opinion about the movie?

- A The movie's storyline is excellent.
- B The movie has a poor implication in symbolism.
- C The movie is perfect in all aspects.
- D The movie is good but could be better.

Communicating Conflict!

Section A

As far back as Hippocrates's time (460-370 B.C.) people have tried to understand other people by characterizing them according to personality type or temperament. Hippocrates believed there were four different body fluids that influenced four basic types of temperament. His work was further developed 500 years later by Galen. These days there are any number of self-assessment tools that relate to the basic descriptions developed by Galen, although we no longer believe the source to be the types of body fluid that dominate our systems.

Section B

The values in self-assessments that help determine personality style. Learning styles, communication styles, conflict-handling styles, or other aspects of individuals is that they help depersonalize conflict in interpersonal relationships. The depersonalization occurs when you realize that others aren't trying to be difficult, but they need different or more information than you do. They're not intend to be rude: they are so focused on the task they forget about greeting people. They would like to work faster but not at the risk of damaging the relationships needed to get the job done. They understand there is a job to do. But it can only be done right with the appropriate information, which takes time to collect. When used appropriately, understanding communication styles can help resolve conflict on teams. Very rarely are conflicts true personality issues. Usually they are issues of style, information needs, or focus.

Section C

Hippocrates and later Galen determined there were four basic temperaments: sanguine, phlegmatic, melancholic and choleric. These descriptions were developed centuries ago and are still somewhat apt, although you could update the wording. In today's world, they translate into the four fairly common communication styles described below:

Section D

The sanguine person would be the expressive or spirited style of communication. These people speak in pictures. They invest a lot of emotion and energy in their communication and often speak quickly. Putting their whole body into it. They are easily sidetracked onto a story that may or may not illustrate the point they are trying to make. Because of their enthusiasm, they are great team motivators. They are concerned about people and relationships. Their high levels of energy can come on strong at times and their focus is usually on the bigger picture, which means they sometimes miss the details or the proper order of things. These people find conflict or differences of opinion invigorating and love to engage in a spirited discussion. They love change and are constantly looking for new and exciting adventures.

Section E

The phlegmatic person – cool and persevering – translates into the technical or systematic communication style. This style of communication is focused on facts and technical details. Phlegmatic people have an orderly methodical way of approaching tasks, and their focus is very much on the task, not on the people, emotions, or concerns that the task may evoke. The focus is also more on the details necessary to accomplish a task.

Sometimes the details overwhelm the big picture and focus needs to be brought back to the context of the task. People with this style think the facts should speak for themselves, and they are not as comfortable with conflict. They need time to adapt to change and need to understand both the logic of it and the steps involved.

Section F

The melancholic person who is softhearted and oriented toward doing things for others translates into the considerate or sympathetic communication style. A person with this communication style is focused on people and relationships. They are good listeners and do things for other people – sometimes to the detriment of getting things done for themselves. They want to solicit everyone's opinion and make sure everyone is comfortable with whatever is required to get the job done. At times this focus on others can distract from the task at hand. Because they are so concerned with the needs of others and smooth over issues, they do not like conflict. They believe that change threatens the status quo and tends to make people feel uneasy, so people with this communication style, like phlegmatic people need time to consider the changes in order to adapt to them.

Section G

The choleric temperament translates into the bold or direct style of communication. People with this style are brief in their communication – the fewer words the better. They are big-picture thinkers and love to be involved in many things at once. They are focused on tasks and outcomes and often forget that the people involved in carrying out the tasks have needs. They don't do detailed work easily and as a result can often underestimate how much time it takes to achieve the task. Because they are so direct, they often seem forceful and can be very intimidating to others. They usually would welcome someone challenging them. But most other styles are afraid to do so. They also thrive on change, the more the better.

Section H

A well-functioning team should have all of these communication styles for true effectiveness. All teams need to focus on the task, and they need to take care of relationships in order to achieve those tasks. They need the big picture perspective or the context of their work, and they need the details to be identified and taken care of for success. We all have aspects of each style within us. Some of us can easily move from one style to another and adapt our style to the needs of the situation at hand- whether the focus is on tasks or relationships. For others, a dominant style is very evident, and it is more challenging to see the situation from the perspective of another style.

The work environment can influence communication styles either by the type of work that is required or by the predominance of one style reflected in that environment. Some people use one style at work and another at home. The good news about communication styles is that we have the ability to develop flexibility in our styles. The greater the flexibility we have, the more skilled we usually are at handling possible and actual conflicts. Usually, it has to be relevant to us to do so, either because we think it is important or because there are incentives in our environment to encourage it. The key is that we have to want to become flexible with our communication style. As Henry Ford said, "Whether you think you can or you can't, you're right!"

Questions 14-21

Reading Passage 2 has eight sections **A-H**

Choose the correct heading for each section from the list of headings below.

Write the correct number, i-x, in boxes 14-21 on your answer sheet.

List of Headings

- i. Different personality types mentioned
- ii. recommendation of combined styles for group
- iii. Historical explanation of understanding personality
- iv. A lively and positive attitude person depicted
- v. A personality likes challenge and direct communication
- vi. different characters illustrated
- vii. Functions of understanding communication styles
- viii. Cautious and considerable person cited
- ix. Calm and Factual personality illustrated
- x. Self-assessment determines one's temperament

14 Section A

15 Section B

16 Section C

17 Section D

18 Section E

19 Section F

20 Section G

21 Section H

Questions 22-25

Do the following statements agree with the information given in Reading Passage 2

In boxes 22-25 on your answer sheet, write

TRUE if the statement is true

FALSE if the statement is false

NOT GIVEN if the information is not given in the passage

22 It is believed that sanguine people do not like a variety

23 Melancholic and phlegmatic people have similar characteristics

24 It is the sanguine personality that is needed most in the workplace.

25 It is possible for someone to change the type of personality.

26 Work surroundings can affect which communication style is the most effective.

Question 27

Choose the correct letter **A, B, C or D**.

Write your answers in box 27 on your answer sheet.

27 The author thinks self-assessment tools can be able to

A assist in developing one's personality in a certain scenario.

B help to understand colleagues and resolve problems

C improve relationship with boss of company

D change others behaviour and personality

Memory and Age

A Aging, it is now clear, is part of an ongoing maturation process that all our organs go through. "In a sense, aging is keyed to the level of vigor of the body and the continuous interaction between levels of body activity and levels of mental activity," reports Arnold B. Scheibel, M.D., whose very academic title reflects how once far-flung domains now converge on the mind and the brain. Scheibel is a professor of anatomy, cell biology, psychiatry, and behavioral sciences at the University of California at Los Angeles, and director of the university's Brain Research Institute. Experimental evidence has backed up popular assumptions that the aging mind undergoes decay analogous to that of the aging body. Younger monkeys, chimps, and lower animals consistently outperform their older colleagues on memory tests. In humans, psychologists concluded, memory and other mental functions deteriorate over time because of inevitable organic changes in the brain as neurons die off. The mental decline after young adulthood appeared inevitable.

B Equipped with imaging techniques that capture the brain in action, Stanley Rapoport, Ph.D., at the National Institutes of Health, measured the flow of blood in the brains of old and young people as they went through the task of matching photos of faces. Since blood flow reflects neuronal activity, Rapoport could be compared with networks of neurons were being used by different subjects. "Even when the reaction times of older and younger subjects were the same, the neural networks they used were significantly different. The older subjects were using different internal strategies to accomplish the same result in the same time," Rapoport says. Either the task required greater effort on the part of the older subjects or the work of neurons originally involved in tasks of that type had been taken over by other neurons, creating different networks.

C At the Georgia Institute of Technology, psychologist Timothy Salthouse, Ph.D., compared a group of very fast and accurate typists of college age with another group in their 60s. since reaction time is faster in younger people and most people's fingers grow less nimble with age, younger typists might be expected to tap right along while the older ones fumble. But both typed 60 words a minute. The older typists, it turned out, achieved their speed with cunning little strategies that made them far more efficient than their younger counterparts: They made fewer finger movements, saving a fraction of a second here and there. They also read ahead in the text. The neural networks involved in typing appear to have been reshaped to compensate for losses in motor skills or other age changes.

D “When a rat is kept in isolation without playmates or objects to interact with, the animal’s brain shrinks, but if we put that rat with 11 other rats in a large cage and give them an assortment of wheels, ladders, and other toys, we can show—after four days— significant differences in its brain,” says Diamond, professor of integrative biology. Proliferating dendrites first appear in the visual association areas. After a month in the enriched environment, the whole cerebral cortex has expanded, and has its blood supply. Even in the enriched environment, rats get bored unless the toys are varied. “Animals are just like we are. They need stimulation,” says Diamond.

One of the most profoundly important mental functions is memory—notorious for its failure with age. So important is memory that the Charles A. Dana foundation recently spent \$8.4 million to set up a consortium of leading medical centers to measure memory loss and aging through brain-imaging technology, neurochemical experiments, and cognitive and psychological tests. One thing, however, is already fairly clear—many aspects of memory are not a function of age at all but of education. Memory exists in more than one form. What we call knowledge—facts—is what psychologists such as Harry P. Bahrick, Ph.D., of Ohio Wesleyan University calls semantic memory. Events, conversations, and occurrences in time and space, on the other hand, make up episodic or event memory, which is triggered by cues from the context. If you were around in 1963 you don’t need to be reminded of the circumstances surrounding the moment you heard that JFK had been assassinated. That event is etched into your episodic memory.

E When you forget a less vivid item, like buying a roll of paper towels at the supermarket, you may blame it on your aging memory. It’s true that episodic memory begins to decline when most people are in their 50s, but it’s never perfect at any age. “Every memory begins as an event,” says Bahrick. “Through repetition, certain events leave behind a residue of knowledge, or semantic memory. On a specific day in the past, somebody taught you that two and two are four, but you’ve been over that information so often you don’t remember where you learned it. What started as an episodic memory has become a permanent part of your knowledge base.” You remember the content, not the context. Our language knowledge, our knowledge of the world and of people, is largely that permanent or semi-permanent residue.

F Probing the longevity of knowledge, Bahrick tested 1,000 high school graduates to see how well they recalled their algebra. Some had completed the course as recently as a month before, others as long as 50 years earlier. He also determined how long each person had studied algebra, the grade received, and how much the skill was used over the course of adulthood. Surprisingly, a person’s grasp of algebra at the time of testing did not depend on how long ago he’d taken the course—the determining factor was the duration of instruction. Those who had spent only a few months learning algebra forgot most of it within two or three years.

G In another study, Bahrick discovered that people who had taken several courses in Spanish, spread out over a couple of years, could recall, decades later, 60 percent or more of the vocabulary they learned. Those who took just one course retained only a trace after three years. “This long-term residue of knowledge remains stable over the decades, independent of the age of the person and the age of the memory. No serious deficit appears until people get to their 50s and 60s, probably due to the degenerative processes of aging rather than a cognitive loss.”

H “You could say metamemory is a byproduct of going to school,” says psychologist Robert Kail, Ph.D., of Purdue University, who studies children from birth to 20 years, the time of life when mental development is most rapid. “The question-and-answer process, especially exam-taking, helps children learn—and also teaches them how their memory works. This may be one reason why, according to a broad range of studies in people over 60, the better educated a person is, the more likely they are to perform better in life and on psychological tests. A group of adult

novice chess players were compared with a group of child experts at the game. In tests of their ability to remember a random series of numbers, the adults, as expected, outscored the children. But when asked to remember the patterns of chess pieces arranged on a board, the children won. "Because they'd played a lot of chess, their knowledge of chess was better organized than that of the adults, and their existing knowledge of chess served as a framework for new memory," explains Kail.

I Specialized knowledge is a mental resource that only improves with time. Crystallized intelligence about one's occupation apparently does not decline at all until at least age 75, and if there is no disease or dementia, may remain even longer. Special knowledge is often organized by a process called "chunking." If procedure A and procedure B are always done together, for example, the mind may merge them into a single command. When you apply yourself to a specific interest—say, cooking—you build increasingly elaborate knowledge structures that let you do more and do it better. This ability, which is tied to experience, is the essence of expertise. Vocabulary is one such specialized form of accrued knowledge. Research clearly shows that vocabulary improves with time. Retired professionals, especially teachers and journalists, consistently score higher on tests of vocabulary and general information than college students, who are supposed to be in their mental prime.

Questions 14-17

Choose the correct letter **A, B, C or D**.

Write your answers in boxes **14-17** on your answer sheet.

14 What does the experiment of the typist show in the passage?

- A Old people reading ability is superior.
- B Losses of age is irreversible.
- C Seasoned tactics made elders more efficient.
- D Old people performed poorly in driving test.

15 Which is correct about rat experiment?

- A Different toys have different effect for rats.
- B Rat's brain weight increased in both cages.
- C Isolated rat's brain grows new connections
- D Boring and complicated surroundings affect brain development.

16 What can be concluded in chess game of children group?

- A They won game with adults.
- B Their organization of chess knowledge is better.
- C Their image memory is better than adults.
- D They used different part of brain when playing chess.

17 What is author's purpose of using "vocabulary study" at the end of passage?

- A Certain people are sensitive to vocabularies while others aren't
- B Teachers and professionals won by their experience
- C Vocabulary memory as a crystallized intelligence is hard to decline
- D Old people use their special zone of brain when study

Questions 18-23

Complete the following summary of the paragraphs of Reading Passage

Using **NO MORE THAN TWO WORDS** from the Reading Passage for each answer.

Write your answers in boxes **18-23** on your answer sheet.

It's long been known that as one significant mental function, **18** deteriorates with age. Charles A. Dana foundation invested millions of dollars to test memory decline. They used advanced technology,

neurochemical experiments and ran several cognitive and **19**..... experiments. Bahrick called one form “**20**.....”, which describes factual knowledge. Another one called “**21**.....” contains events in time and space format. He conducted two experiments toward to knowledge memory’s longevity, he asked 1000 candidates some knowledge of **22**....., some could even remember it decades ago. Second research of Spanish course found that multiple courses participants could remember more than half of **23**..... They learned after decades, whereas single course taker only remembered as short as 3 years.

Questions 24-27

Use the information in the passage to match the people (listed **A-F**) with opinions or deeds below.

Write the appropriate letters A-F in boxes 24-27 on your answer sheet.

- A** Harry P. Bahrick
- B** Arnold B. Scheibel
- C** Marion Diamond
- D** Timothy Salthouse
- E** Stanley Rapport
- F** Robert Kail

24 Examined both young and old’s blood circulation of brain while testing.

25 Aging is a significant link between physical and mental activity.

26 Some semantic memory of a event would not fade away after repetition.

27 Rat’s brain developed when put in a diverse environment.

Seed Hunting

A With quarter of the world’s plants set to vanish within the next 50 years, Dough Alexander reports on the scientists working against the clock the preserve the Earth’s botanical heritage. They travel the four corners of the globe, scouring jungles, forests and savannas. But they’re not looking for ancient artifacts, lost treasure or undiscovered tombs. Just pods. It may lack the romantic allure of archaeology, or the whiff of danger that accompanies going after big game, but seed hunting is an increasingly serious business. Some seek seeds for profit-hunters in the employ of biotechnology firms, pharmaceutical companies and private corporations on the lookout for species that will yield the drugs or crops of the future. Others collect to conserve, working to halt the sad slide into extinction facing so many plant species.

B Among the pioneers of this botanical treasure hunt was John Tradescant, an English royal gardener who brought back plants and seeds from his journeys abroad in the early 1600s. Later, the English botanist Sir Joseph Banks – who was the first director of the Royal Botanic Gardens at Kew and travelled with Captain James Cook on his voyages near the end of the 18th century – was so driven to expand his collections that he sent botanists around the world at his own expense.

C Those heady days of exploration and discovery may be over, but they have been replaced by a pressing need to preserve our natural history for the future. This modern mission drives hunters such as Dr Michiel van Slageren, a good-natured Dutchman who often sports a wide-brimmed hat in the field – he could easily be mistaken for the cinematic hero Indiana Jones. He and three other seed hunters work at the Millennium Seed Bank, an 80 million [pounds sterling] international conservation project that aims to protect the world’s most endangered wild plant species.

D The group’s headquarters are in a modern glass-and-concrete structure on a 200- hectare Estate at Wakehurst

Place in the West Sussex countryside. Within its underground vaults are 260 million dried seeds from 122 countries, all stored at -20 Celsius to survive for centuries. Among the 5,100 species represented are virtually all of Britain's 1,400 native seed-bearing plants, the most complete such collection of any country's flora.

E Overseen by the Royal Botanic Gardens, the Millennium Seed Bank is the world's largest wild-plant depository. It aims to collect 24,000 species by 2010. The reason is simple: thanks to humanity's effort, an estimated 25 percent of the world's plants are on the verge of extinction and may vanish within 50 years. We're currently responsible for habitat destruction on an unprecedented scale, and during the past 400 years, plant species extinction rates have been about 70 times greater than those indicated by the geological record as being 'normal'. Experts predict that during the next 50 years a further one billion hectares of wilderness will be converted to farmland in developing countries alone.

F The implications of this loss are enormous. Besides providing staple food crops, plants are a source of many medicines and the principal supply of fuel and building materials in many parts of the world. They also protect the soil and help regulate the climate. Yet, across the globe, plant species are being driven to extinction before their potential benefits are discovered.

G The world Conservation Union has listed 5,714 threatened species is sure to be much higher. In the UK alone, 300 wild plant species are classified as endangered. The Millennium Seed Bank aims to ensure that even if a plant becomes extinct in the wild, it won't be lost forever. Stored seeds can be used to help restore damaged or destroyed environments or in scientific research to find new benefits for society- in medicine, agriculture or local industry- that would otherwise be lost.

H Seed banks are an insurance policy to protect the world's plant heritage for the future, explains Dr Paul Smith, another Kew seed hunter. "Seed conservation techniques were originally developed by farmers," he says. "Storage is the basis what we do, conserving seeds until you can use them just as in farming." Smith says there's no reason why any plant species should become extinct, given today's technology. But he admits that the biggest challenge is finding, naming and categorizing all the world's plants. And someone has to gather these seeds before it's too late. "There aren't a lot of people out there doing this," he says. "The key is to know the flora from a particular area, and that knowledge takes years to acquire."

I There are about 1,470 seedbanks scattered around the globe, with a combined total of 5.4 million samples, of which perhaps two million are distinct non-duplicates. Most preserve genetic material for agricultural use in order to ensure crop diversity; others aim to conserve wild species, although only 15 percent of all banked plants are wild.

J Many seed banks are themselves under threat due to a lack of funds. Last year, Imperial College, London, examined crop collections from 151 countries and found that while the number of plant samples had increased in two-thirds of the countries, the budget had been cut in a quarter and remained static in another 35 percent. The UN's Food and Agriculture Organization and the Consultative Group on International Agricultural Research has since set up the Global Conservation Trust, which aims to raise US \$260 million to protect seed banks in perpetuity.

Questions 14-19

Do the following statements agree with the information given in Reading Passage 2?

In boxes **14-19** on your answer sheet, write

TRUE if the statement is true

FALSE if the statement is false

NOT GIVEN if the information is not given in the passage

14 The purpose of collecting seeds now is different from the past.

15 The millennium seed bank is the earliest seed bank.

16 One of major threats for plant species extinction is farmland expansion into wildness.

17 The approach that scientists apply to store seeds is similar to that used by farmers.

18 Technological development is the only hope to save plant species.

19 The works of seed conservation are often limited by financial problems.

Questions 20-24

Complete the following summary of the paragraphs of Reading Passage 2,

Using **NO MORE THAN THREE WORDS** from the Reading Passage for each answer.

Write your answers in boxes **20-24** on your answer sheet.

Some people collect seeds for the purpose of protecting certain species from **20**.....; others collect seeds for their ability to produce **21** They are called seed hunters. The **22**.....Of them included both gardeners and botanists, such as **23**....., who financially supported collectors out of his own pocket. The seeds collected are usually stored in seed banks, one of which is the famous millennium seed bank, where seeds are all stored in the **24**.....at a low temperature.

Questions 25-26

Choose the correct letter, **A-E**.

Write your answers in boxes **25, 26** on your answer sheet.

Which **TWO** of the following are provided by plants to the human?

- A** food
- B** fuels
- C** clothes
- D** energy
- E** commercial products

Can Scientists Tell Us: What happiness is?

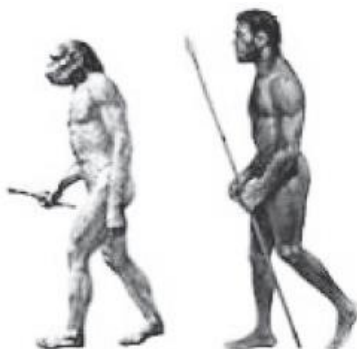
A Economists accept that if people describe themselves as happy, then they are happy. However, psychologists differentiate between levels of happiness. The most immediate type involves a feeling —pleasure or joy. But sometimes happiness is a judgment that life is satisfying, and does not imply an emotional state. Esteemed psychologist Martin Seligman has spearheaded an effort to study the science of happiness. The bad news is that we're not wired to be happy. The good news is that we can do something about it. Since its origins in a Leipzig laboratory 130 years ago, psychology has had little to say about goodness and contentment. Mostly psychologists have concerned themselves with weakness and misery. There are libraries full of theories about why we get sad, worried, and angry. It hasn't been respectable science to study what happens when lives go well. Positive experiences, such as joy, kindness, altruism and heroism, have mainly been ignored. For every 100 psychology papers dealing with anxiety or depression, only one concerns a positive trait.



B A few pioneers in experimental psychology bucked the trend. Professor Alicelsen of Cornell University and colleagues have demonstrated how positive emotions make people think faster and more creatively. Showing how easy it is to give people an intellectual boost, Isen divided doctors making a tricky diagnosis into three groups: one received candy, one read humanistic statements about medicine, one was a control group. The doctors who had candy displayed the most creative thinking and worked more efficiently. Inspired by Isen and others, Seligman got stuck in. He raised millions of dollars of research money and funded 50 research groups involving 150 scientists across the world. Four positive psychology centres opened, decorated in cheerful colours and furnished with sofas and babysitters. There were get-togethers on Mexican beaches where psychologists would snorkel and eat fajitas, then form 'pods' to discuss subjects such as wonder and awe. A thousand therapists were coached in the new science.

C But critics are demanding answers to big questions. What is the point of defining levels of happiness and classifying the virtues? Aren't these concepts vague and impossible to pin down? Can you justify spending funds to research positive states when there are problems such as famine, flood and epidemic depression to be solved? Seligman knows his work can be belittled alongside trite notions such as 'the power of positive thinking'. His plan to stop the new science floating 'on the waves of self-improvement fashions' is to make sure it is anchored to positive philosophy above, and to positive biology below.

D And this takes us back to our evolutionary past. Homo sapiens evolved during the Pleistocene era (1.8 m to 10,000 years ago), a time of hardship and turmoil. It was the Ice Age, and our ancestors endured long freezes as glaciers formed, then ferocious floods as the ice masses melted. We shared the planet with terrifying creatures such as mammoths, elephant-sized ground sloths and sabre toothed cats. But by the end of the Pleistocene, all these animals were extinct. Humans, on the other hand, had evolved large brains and used their intelligence to make fire and sophisticated tools, to develop talk and social rituals. Survival in a time of adversity forged our brains into a persistent mould. Professor Seligman says: 'Because our brain evolved during a time of ice, flood and famine, we have a catastrophic brain. The way the brain works is looking for what's wrong. The problem is, that worked in the Pleistocene era. It favoured you, but it doesn't work in the modern world.'



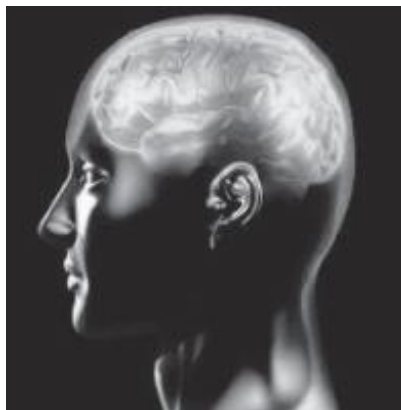
E Although most people rate themselves as happy, there is a wealth of evidence to show that negative thinking is deeply ingrained in the human psyche. Experiments show that we remember failures more vividly than successes. We dwell on what went badly, not what went well. Of the six universal emotions, four anger, fear, disgust and sadness are negative and only one, joy, is positive. (The sixth, surprise, is neutral. According to the psychologist Daniel Nettle, author of *Happiness*, and one of the Royal Institution lecturers, the negative emotions each tell us 'something bad has happened' and suggest a different course of action.

F What is it about the structure of the brain that underlies our bias towards negative thinking? And is there a

biology of joy? At Iowa University, neuroscientists studied what happens when people are shown pleasant and unpleasant pictures. When subjects see landscapes or dolphins playing, part of the frontal lobe of the brain becomes active. But when they are shown unpleasant images a bird covered in oil, or a dead soldier with part of his face missing, the response comes from more primitive parts of the brain. The ability to feel negative emotions derives from an ancient danger-recognition system formed early in the brain's evolution. The pre-frontal cortex, which registers happiness, is the part used for higher thinking, an area that evolved later in human history.



G Our difficulty, according to Daniel Nettle, is that the brain systems for liking and wanting are separate. Wanting involves two ancient regions the amygdala and the nucleus accumbens that communicate using the chemical dopamine to form the brain's reward system. They are involved in anticipating the pleasure of eating and in addiction to drugs. A rat will press a bar repeatedly, ignoring sexually available partners, to receive electrical stimulation of the 'wanting' parts of the brain. But having received brain stimulation, the rat eats more but shows no sign of enjoying the food it craved. In humans, a drug like nicotine produces much craving but little pleasure.



H In essence, what the biology lesson tells us is that negative emotions are fundamental to the human condition, and it's no wonder they are difficult to eradicate. At the same time, by a trick of nature, our brains are designed to crave but never really achieve lasting happiness.

Questions 14-20

The Reading Passage has eight paragraphs **A-H**.

Which paragraph contains the following information?

Write the correct letter **A-H**, in boxes **14-20** on your answer sheet.

14 An experiment involving dividing several groups one of which received a positive icon

15 Review of a poorly researched psychology area

- 16 Contrast being made about the brains' action as response to positive or negative stimulus
- 17 The skeptical attitude that the research seemed to be a waste of fund
- 18 a substance that produces much wanting instead of much liking
- 19 a conclusion that lasting happiness are hardly obtained because of the nature of brains
- 20 One description that listed the human emotional categories.

Questions 21-25

Complete the following summary of the paragraphs of Reading Passage, using **NO MORE THAN FOUR WORDS** from the passage for each answer. Write your answers in boxes **21-25** on your answer sheet.

A few pioneers in experimental psychology study what happens when lives go well. Professor Alice divided doctors, making a tricky experiment, into three groups: besides the one control group, the other two either are asked to read humanistic statements about drugs, or received **21** _____. The latter displayed the most creative thinking and worked more efficiently. Since critics are questioning the significance of the **22** _____ for both levels of happiness and classification for the virtues. Professor Seligman countered in an evolutionary theory: survival in a time of adversity forged our brains into the way of thinking for what's wrong because we have a **23** _____.

There is bountiful of evidence to show that negative thinking is deeply built in the human psyche. Later, at Iowa University, neuroscientists studied the active parts in brains to contrast when people are shown pleasant and unpleasant pictures. When positive images like **24** _____ are shown, part of the frontal lobe of the brain becomes active. But when they are shown unpleasant image, the response comes from **25** _____ of the brain.

Questions 26

Choose the correct letter, A, B, C or D.

Write your answers in boxes **26** on your answer sheet.

According to Daniel Nettle in the last two paragraphs, what is true as the scientists can tell us about happiness?

- A Brain systems always mix liking and wanting together.
- B Negative emotions can be easily get rid of if we think positively.
- C Happiness is like nicotine we are craving for but get little pleasure.
- D The inner mechanism of human brains does not assist us to achieve durable happiness.

Questions 14-20

READING PASSAGE 2 has seven paragraphs, **A-G**.

Choose the correct heading for each paragraph from the list of headings below.

Write the correct number, **i-x**, in boxes **14-20** on your answer sheet.

List of headings

- i The best moment to migrate
- ii The unexplained rejection of closer feeding ground
- iii The influence of weather on the migration route
- iv Physical characteristics that allow birds to migrate
- v The main reason why birds migrate
- vi The best wintering grounds for birds
- vii Research findings on how birds migrate
- viii Successful migration despite trouble of wind
- ix Contrast between long-distance migration and short-distance migration
- x Mysterious migration despite lack of teaching

- 14 Paragraph A
- 15 Paragraph B
- 16 Paragraph C
- 17 Paragraph D
- 18 Paragraph E
- 19 Paragraph F
- 20 Paragraph G

Bird Migration

A Birds have many unique design features that enable them to perform such amazing feats of endurance. They are equipped with lightweight, hollow bones, intricately designed feathers providing both lift and thrust for rapid flight, navigation systems superior to any that man has developed, and an ingenious heat-conserving design that, among other things, concentrates all blood circulation beneath layers of warm, waterproof plumage, leaving them fit to face life in the harshest of climates. Their respiratory systems have to perform efficiently during sustained flights at altitude, so they have a system of extracting oxygen from their lungs that far exceeds that of any other animal. During the later stages of the summer breeding season, when food is plentiful, their bodies are able to accumulate considerable layers of fat, in order to provide sufficient energy for their long migratory flights.

B The fundamental reason that birds migrate is to find adequate food during the winter months when it is in short supply. This particularly applies to birds that breed in the temperate and Arctic regions of the Northern Hemisphere, where food is abundant during the short growing season. Many species can tolerate cold temperatures if food is plentiful, but when food is not available they must migrate. However, intriguing questions remain.

C One puzzling fact is that many birds journey much further than would be necessary just to find food and good weather. Nobody knows, for instance, why British swallows, which could presumably survive equally well if they spent the winter in equatorial Africa, instead fly several thousands of miles further to their preferred winter home in South Africa's Cape Province. Another mystery involves the huge migrations performed by arctic terns and mudflat-feeding shorebirds that breed close to Polar Regions. In general, the further north a migrant species breeds, the further south it spends the winter. For arctic terns this necessitates an annual round trip of 25,000 miles. Yet, en route to their final destination in far-flung southern latitudes, all these individuals overfly other areas of seemingly suitable habitat spanning two hemispheres. While we may not fully understand birds' reasons for going to particular places, we can marvel at their feats.

D One of the greatest mysteries is how young birds know how to find the traditional wintering areas without parental guidance. Very few adults migrate with juveniles in tow, and youngsters may even have little or no inkling of their parents' appearance. A familiar example is that of the cuckoo, which lays its eggs in another species nest and never encounters its young again. It is mind-boggling to consider that, once raised by its host species, the young cuckoo makes its own way to ancestral wintering grounds in the tropics before returning single-handedly to northern Europe the next season to seek out a mate among its own kind. The obvious implication is that it inherits from its parents an inbuilt route map and direction-finding capability, as well as a mental image of what another cuckoo looks like. Yet nobody has the slightest idea as to how this is possible.

E Mounting evidence has confirmed that birds use the positions of the sun and stars to obtain compass directions. They seem also to be able to detect the earth's magnetic field, probably due to having minute crystals of magnetite in the region of their brains. However, true navigation also requires an awareness of position and time, especially when lost. Experiments have shown that after being taken thousands of miles over an unfamiliar landmass, birds are still capable of returning rapidly to nest sites. Such phenomenal powers are the product of

computing a number of sophisticated cues, including an inborn map of the night sky and the pull of the earth's magnetic field. How the birds use their 'instruments' remains unknown, but one thing is clear: they see the world with a superior sensory perception to ours. Most small birds migrate at night and take their direction from the position of the setting sun. However, as well as seeing the sun go down, they also seem to see the plane of polarized light caused by it, which calibrates their compass. Traveling at night provides other benefits. Daytime predators are avoided and the danger of dehydration due to flying for long periods in warm, sunlit skies is reduced. Furthermore, at night the air is generally cooler and less turbulent and so conducive to sustained, stable flight.

F Nevertheless, all journeys involve considerable risk, and part of the skill in arriving safely is setting off at the right time. This means accurate weather forecasting, and utilizing favorable winds. Birds are adept at both, and, in laboratory tests, some have been shown to detect the minute difference in barometric pressure between the floor and ceiling of a room. Often birds react to weather changes before there is any visible sign of them. Lapwings, which feed on grassland, flee west from the Netherlands to the British Isles, France and Spain at the onset of a cold snap. When the ground surface freezes the birds could starve. Yet they return to Holland ahead of a thaw, their arrival linked to a pressure change presaging an improvement in the weather.

G In one instance a Welsh Manx shearwater carried to America and released was back in its burrow on Skokholm Island, off the Pembrokeshire coast, one day before a letter announcing its release! Conversely, each autumn a small number of North American birds are blown across the Atlantic by fast-moving westerly tailwinds. Not only do they arrive safely in Europe, but, based on ringing evidence, some make it back to North America the following spring, after probably spending the winter with European migrants in sunny African climes.

Questions 21-22

Choose **TWO** letters, **A-E**.

Write the correct letters in boxes **21** and **22** on your answer sheet.

Which **TWO** of the following statements are true of bird migration?

- A Birds often fly further than they need to.
- B Birds traveling in family groups are safe.
- C Birds flying at night need less water.
- D Birds have much sharper eye-sight than humans.
- E Only shorebirds are resistant to strong winds.

Question 23-26

Complete the sentences below using **NO MORE THAN TWO WORD OR NUMBER** from the passage.

Write your answers in boxes **23-26** on your answer sheet.

- 23** It is a great mystery that young birds like cuckoos can find their wintering grounds without _____.
- 24** Evidence shows birds can tell directions like a _____ by observing the sun and the stars.
- 25** One advantage for birds flying at night is that they can avoid contact with _____.
- 26** Laboratory tests show that birds can detect weather without _____ signs.

Facial Expressions

A Facial expressions are a form of non-verbal communication, which make up a large part of our daily communication. By facial movements, we can suggest volumes of information without 'speaking any words'. They are a primary means of conveying social information among humans but are also found in most other mammals and a few other animal species. Facial expressions and their significance in the perceiver can, to some extent, vary between cultures with evidence from descriptions in the works of Charles Darwin. Understanding facial impressions is of great importance, and also, some essential elements should be taken into account when people interpret the impressions.

B Facial expressions are closely tied to emotions. Expressions are configurations of different small muscle movements in the face that show an individual's discrete emotional states and a vast amount of cognitive information. A number of studies have proved that facial expressions are viewed as one of the best measures of emotional response. Sometimes words do not necessarily coincide with emotions, but the face might betray a person's actual feelings. At the same time, the close link between emotions and expressions can also work in the other direction; it has been observed that voluntarily assuming an expression can actually cause the associated emotion.

C Some expressions like anger and extreme contentment can be accurately comprehended even between members of different species. Others, however, are hard to interpret even in familiar individuals. For instance, disgust and fear can be tough to distinguish. Because faces have only a limited range of movements, expressions rely upon fairly slight differences in the proportions and relative positions of facial features, and reading them requires considerable sensitivity to some. Meanwhile, one's life experiences, beliefs and conceptual knowledge will have an influence on how he deciphers the facial expressions.

D Also, a person's eyes reveal much about how they are feeling or what they are thinking. Research by Boston College professor Joe Tecce suggests that stress levels are reflected by blink rates. He supports his findings with statistics on the relation between the blink rates of presidential candidates and their success in their races. Tecce claims that the faster blinker in the presidential debates has lost every election since 1980, which seems interesting. However, it is important to recognise that non-verbal communication is multichannelled, and focusing on only one aspect is biased. Nervousness can also be measured by examining each candidate's perspiration, eye contact and stiffness.

E As Charles Darwin, the first person who pointed out that the facial expressions of particular emotions were universal, noted in his book *The Expression of the Emotions in Man and Animals*: 'The young and the old of widely different races, both with man and animals, express the same state of mind by the same muscle movement.' Still, up to the mid-20th century, most anthropologists believed that facial expressions were entirely learned and could therefore differ among cultures. Studies conducted in the 1960s by Paul Ekman eventually supported Darwin's belief to a large degree.

F Ekman's opinion, which was contrary to that of some anthropologists including Margaret Mead, showed that facial expressions of emotions are not culturally determined but universal across human cultures. During the period between 1967 and 1968, Dr. Ekman travelled to Papua New Guinea to investigate the non-verbal communication behaviours of the South Fore people. If the South Fore tribe, a remote and primitive one, could interpret the facial expressions as their western counterparts, we would gain significant evidence of the universality of facial expressions. Subjects, including 189 adults and 130 children from a very isolated population, as well as 23 members who lived in a less-isolated lifestyle, were shown three pictures (two for children) of facial expressions and then asked to match those pictures to three different scenarios that described particular emotions.

G It was discovered that the isolated South Fore people could identify emotions with the same accuracy as the less-isolated control group, but the study displayed some problems including the fact that both fear and surprise are constantly misidentified. He also drew a conclusion that certain facial expressions correspond to particular emotions and cannot be covered, no matter what the cultural background is, and regardless of whether or not the

culture has been isolated or exposed to the mainstream.

H Expressions Ekman found to be generally recognised included those indicating anger, disgust, fear, joy, sadness and surprise (note that none of these emotions has a definitive social component, such as shame, pride or schadenfreude). Findings on contempt (which is social), an expression which can occur with or without a hint of small muscle motions, are less clear, though there is at least some preliminary evidence that this emotion and its expression are universal. This may suggest that the facial expressions are largely related to the mind, with each part of the face expressing a particular emotion.

Questions 14 - 19

Reading Passage 2 has eight paragraphs, **A–H**.

Which paragraph contains the following information?

Write the correct letter, **A–H**, in boxes **14–19** on your answer sheet.

NB You may use any letter more than once.

14 mention of the data correlation between blink frequency and election results

15 early records of how culture influences the interpretation of facial expressions

16 the difficulty in identifying the actual meaning of facial expressions

17 reference to the sociality of several facial expressions

18 various indicators of how nervous a person is

19 mention of the relation between emotions and facial expressions

Questions 20 - 24

Complete the summary below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Write your answers in boxes **20–24** on your answer sheet.

Ekman's findings demonstrated that fear and surprise are persistently **20** _____, and Ekman concluded that some facial expressions coincide with certain **21** _____ and are impossible to be masked, regardless of **22** _____, or whether the culture is **23** _____ or **24** _____ the mainstream.

Questions 25 - 26

Choose **TWO** letters, **A–E**.

Write the correct letters in boxes 25 and 26 on your answer sheet.

25-26 Which **TWO** of the following statements are true according to Ekman's theory?

A There is no evidence that animals have their own facial expressions

B Facial expressions are associated with different cultures

C There is an underlying relationship between facial expressions and mental states

D Men's facial expressions are more controlled by psychology than women's.

E Facial expressions that express certain emotions are the same even across cultures

Leaf-Cutting Ants and Fungus

A The ants and their agriculture have been extensively studied over the years, but the recent research has uncovered intriguing new findings about the fungus they cultivate, how they domesticated it and how they cultivate it and preserve it from pathogens. For example, the fungus farms, which the ants were thought to keep free of pathogens, turn out to be vulnerable to a devastating mold, found nowhere else but in ants' nests. To keep the mold in check, the ants long ago made a discovery that would do credit to any pharmaceutical laboratory.



B Leaf-cutting ants and their fungus farms are a marvel of nature and perhaps the best known example of symbiosis, the mutual dependence of two species. The ants' achievement is remarkable — the biologist Edward O. Wilson has called it 'one of the major breakthroughs in animal evolution' — because it allows them to eat, courtesy of their mushroom's digestive powers, the otherwise poisoned harvest of tropical forests whose leaves are laden with terpenoids, alkaloids and other chemicals designed to sicken browsers.

C Fungus growing seems to have originated only once in evolution, because all gardening ants belong to a single tribe, the descendants of the first fungus farmer. There are more than 200 known species of the attine ant tribe, divided into 12 groups, or genera. The leaf-cutters use fresh vegetation; the other groups, known as the lower attines because their nests are smaller and their techniques more primitive, feed their gardens with detritus like dead leaves, insects and feces.

D The leaf-cutters' fungus was indeed descended from a single strain, propagated clonally, or just by budding, for at least 23 million years. But the lower attine ants used different varieties of the fungus, and in one case a quite separate species, the four biologists discovered. The pure strain of fungus grown by the leaf-cutters, it seemed to Mr. Currie, resembled the monocultures of various human crops, that are very productive for a while and then succumb to some disastrous pathogen, such as the Irish potato blight. Monocultures, which lack the genetic diversity to respond to changing environmental threats, are sitting ducks for parasites. Mr. Currie felt there had to be a parasite in the ant-fungus system. But a century of ant research offered no support for the idea. Textbooks describe how leaf-cutter ants scrupulously weed their gardens of all foreign organisms. People kept telling me, 'You know the ants keep their gardens free of parasites, don't you?' Mr. Currie said of his efforts to find a hidden interloper.

E But after three years of sifting through attine ant gardens, Mr. Currie discovered they are far from free of infections. In last month's issue of the Proceedings of the National Academy of Sciences, he and two colleagues, Dr. Mueller and David Mairoch, isolated several alien organisms, particularly a family of parasitic molds called Escovopsis.

F Escovopsis turns out to be a highly virulent pathogen that can devastate a fungus garden in a couple of days. It blooms like a white cloud, with the garden dimly visible underneath. In a day or two the whole garden is enveloped. 'Other ants won't go near it and the ants associated with the garden just starve to death,' Dr. Rehner said. 'They just seem to give up, except for those that have rescued their larvae.' The deadly mold then turns greenish-brown as it enters its spore-forming stage.

G Evidently the ants usually manage to keep Escovopsis and other parasites under control. But with any lapse in control, or if the ants are removed, Escovopsis will quickly burst forth. Although new leaf-cutter gardens start off free of Escovopsis, within two years some 60 percent become infected. The discovery of Escovopsis's role brings a new level of understanding to the evolution of the attine ants. 'In the last decade, evolutionary biologists have been

increasingly aware of the role of parasites as driving forces in evolution,’ Dr. Schultz said. There is now a possible reason to explain why the lower attine species keep changing the variety of fungus in their mushroom gardens, and occasionally domesticating new ones — to stay one step ahead of the relentless Escovopsis.



H Interestingly, Mr. Currie found that the leaf-cutters had in general fewer alien molds in their gardens than the lower attines, yet they had more Escovopsis infections. It seems that the price they pay for cultivating a pure variety of fungus is a higher risk from Escovopsis. But the leaf-cutters may have little alternative: they cultivate a special variety of fungus which, unlike those grown by the lower attines, produces nutritious swollen tips for the ants to eat.

I Discovery of a third partner in the ant-fungus symbiosis raises the question of how the attine ants, especially the leaf-cutters, keep this dangerous interloper under control. Amazingly enough, Mr. Currie has again provided the answer. ‘People have known for a hundred years that ants have a whitish growth on the cuticle,’ said Dr. Mueller, referring to the insects’ body surface. ‘People would say this is like a cuticular wax. But Cameron was the first one in a hundred years to put these things under a microscope. He saw it was not inert wax. It is alive.’ Mr. Currie discovered a specialized patch on the ants’ cuticle that harbors a particular kind of bacterium, one well known to the pharmaceutical industry, because it is the source of half the antibiotics used in medicine. From each of 22 species of attine ant studied, Mr. Cameron and colleagues isolated a species of Streptomyces bacterium, they reported in Nature in April. The Streptomyces does not have much effect on ordinary laboratory funguses. But it is a potent poisoner of Escovopsis, inhibiting its growth and suppressing spore formation. It also stimulates growth of the ants’ mushroom fungus. The bacterium is carried by virgin queens when they leave to establish new nests, but is not found on male ants, playboys who take no responsibility in nest-making or gardening.

J Because both the leaf-cutters and the lower attines use Streptomyces, the bacterium may have been part of their symbiosis for almost as long as the Escovopsis mold. If so, some Alexander Fleming of an ant discovered antibiotics millions of years before people did. Even now, the ants are accomplishing two feats beyond the powers of human technology. The leaf-cutters are growing a monocultural crop year after year without disaster, and they are using an antibiotic apparently so wisely and prudently that, unlike people, they are not provoking antibiotic resistance in the target pathogen.

Questions 14-19

Use the information in the passage to match the options (listed **A-C**) with activities or features of ants below. Write the appropriate letters **A-C** in boxes **14-19** on your answer sheet.

NB you may use any letter more than once

- A Leaf-cutting ants
- B Lower attines
- C Both

14 Build small nests and live with different foreign fungus.

- 15 Use toxic leaves to feed fungus.
- 16 Raise fungus which don't live with other foreigners.
- 17 Use substances to fight against meconopsis.
- 18 Use dead vegetable to feed fungus.
- 19 Are free of parasites explained previously.

Questions 20-24

The Reading Passage has ten paragraphs **A-J**.

Which paragraph contains the following information?

Write the correct letter **A-J**, in boxes 20-24 on your answer sheet.

- 20 Dangerous outcome of Escovopsis.
- 21 Disadvantage of growing single fungus.
- 22 Comparison of features of two different nests.
- 23 Two achievements made by ants earlier than human.
- 24 Advantage of growing new breed of fungus.

Questions 25-26

Choose the correct letter, **A, B, C** or **D**.

Write your answers in boxes 25-26 on your answer sheet.

- 25 How does author think of Curries opinion?
 - A his viewpoint was verified later.
 - B earlier study has sufficient evidence.
 - C no details mentioned in article.
 - D his opinion was proved to be wrong.
- 26 What did scientists find on the skin of ants under microscope?
 - A some white cloud mold embed in their skin
 - B that Wax is all over their skin.
 - C a substance which is useful to humans.
 - D a substance which suppresses growth of fungus.

Left-handed or Right-handed

A The probability that two right-handed people would have a left-handed child is only about 9.5 percent. The chance rises to 19.5 percent if one parent is a lefty and 26 percent if both parents are left-handed: The preference, however, could also stem from an infant's imitation of his parents. To test genetic influence, starting in the 1970s British biologist Marian Annett of the University of Leicester hypothesized that no single gene determines handedness. Rather, during fetal development, a certain molecular factor helps to strengthen the brain's left hemisphere, which increases the probability that the right hand will be dominant, because the left side of the brain controls the right side of the body, and vice versa. Among the minority of people who lack this factor, handedness develops entirely by chance. Research conducted on twins complicates the theory, however. One in five sets of identical twins involves one right-handed and one left-handed person, despite the fact that their genetic material is the same. Genes, therefore, are not solely responsible for handedness.



B Genetic theory is also undermined by results from Peter Hepper and his team at Queen's University in Belfast, Ireland. In 2004 the psychologists used ultrasound to show that by the 15th week of pregnancy, fetuses already have a preference as to which thumb they suck. In most cases, the preference continued after birth. At 15 weeks, though, the brain does not yet have control over the body's limbs. Hepper speculates that fetuses tend to prefer whichever side of the body is developing quicker and that their movements, in turn, influence the brain's development. Whether this early preference is temporary or holds up throughout development and infancy is unknown. Genetic predetermination is also contradicted by the widespread observation that children do not settle on either their right or left hand until they are two or three years old.

C But even if these correlations were true, they did not explain what actually causes left-handedness. Furthermore, specialization on either side of the body is common among animals. Cats will favor one paw over another when fishing toys out from under the couch. Horses stomp more frequently with one hoof than the other. Certain crabs motion predominantly with the left or right claw. In evolutionary terms, focusing power and dexterity in one limb is more efficient than having to train two, four or even eight limbs equally. Yet for most animals, the preference for one side or the other is seemingly random. The overwhelming dominance of the right hand is associated only with humans. That fact directs attention toward the brain's two hemispheres and perhaps toward language.

D Interest in hemispheres dates back to at least 1836. That year, at a medical conference, French physician Marc Dax reported on an unusual commonality among his patients. During his many years as a country doctor, Dax had encountered more than 40 men and women for whom speech was difficult, the result of some kind of brain damage. What was unique was that every individual suffered damage to the left side of the brain. At the conference, Dax elaborated on his theory, stating that each half of the brain was responsible for certain functions and that the left hemisphere controlled speech. Other experts showed little interest in the Frenchman's ideas. Over time, however, scientists found more and more evidence of people experiencing speech difficulties following injury to the left brain. Patients with damage to the right hemisphere most often displayed disruptions in perception or concentration. Major advancements in understanding the brain's asymmetry were made in the 1960s as a result of so-called split-brain surgery, developed to help patients with epilepsy. During this operation, doctors severed the corpus callosum—the nerve bundle that connects the two hemispheres. The surgical cut also stopped almost all normal communication between the two hemispheres, which offered researchers the opportunity to investigate each side's activity.

E In 1949 neurosurgeon John Wada devised the first test to provide access to the brain's functional organization of language. By injecting an anesthetic into the right or left carotid artery, Wada temporarily paralyzed one side of a healthy brain, enabling him to more closely study the other side's capabilities. Based on this approach, Brenda Milner and the late Theodore Rasmussen of the Montreal Neurological Institute published a major study in 1975 that confirmed the theory that country doctor Dax had formulated nearly 140 years earlier: in 96 percent of right-handed people, language is processed much more intensely in the left hemisphere. The correlation is not as clear in lefties. However, for two-thirds of them, the left hemisphere is still the most active language processor. But for the remaining third either the right side is dominant or both sides work equally, controlling different language functions. That last statistic has slowed the acceptance of the notion that the predominance of right-handedness is driven by left-hemisphere dominance in language processing. It is not at all clear why language control should somehow have dragged the control of body movement with it. Some experts think one reason the left hemisphere reigns over language is because the organs of speech processing—the larynx and tongue—are positioned on the

body's symmetry axis. Because these structures were centered, it may have been unclear, in evolutionary terms, which side of the brain should control them, and it seems unlikely that shared operation would result in smooth motor activity. Language and handedness could have developed preferentially for very different reasons as well. For example, some researchers, including evolutionary psychologist Michael C. Corballis of the University of Auckland in New Zealand, think that the origin of human speech lies in gestures. Gestures predated words and helped language emerge. If the left hemisphere began to dominate speech, it would have dominated gestures, too, and because the left brain controls the right side of the body, the right hand developed more strongly.



F Perhaps we will know more soon. In the meantime, we can revel in what, if any, differences handedness brings to our human talents. Popular wisdom says right-handed, left-brained people excel at logical, analytical thinking. Left-handed, right-brained individuals are thought to possess more creative skills and may be better at combining the functional features emergent in both sides of the brain. Yet some neuroscientists see such claims as pure speculation. Fewer scientists are ready to claim that left-handedness means greater creative potential. Yet lefties are prevalent among artists, composers and the generally acknowledged great political thinkers. Possibly if these individuals are among the lefties whose language abilities are evenly distributed between hemispheres, the intense interplay required could lead to unusual mental capabilities.

G Or perhaps some lefties become highly creative simply because they must be more clever to get by in our right-handed world. This battle, which begins during the very early stages of childhood, may lay the groundwork for exceptional achievements.

Questions 14-18

The Reading Passage has seven paragraphs **A-G**.

Which paragraph contains the following information?

Write the correct letter **A-G**, in boxes 14-18 on your answer sheet.

NB You may use any letter more than once.

- 14** Phenomenon of using one side of their body for animals.
- 15** Statistics on rate of one-handedness born.
- 16** The age when the preference of using one hand is fixed.
- 17** great talents of occupations in the left-handed population.
- 18** Earliest record of researching hemisphere's function

Questions 19-22

Look at the following researchers (Questions **19-22**) and the list of findings below. Match each researcher with the correct finding.

- A** Brenda Milner
- B** Marian Annett
- C** Peter Hepper
- D** Michale Corballis

- 19** Ancient language evolution is connected to body gesture and therefore influences handedness.
- 20** A child handedness is not determined by just biological factors.
- 21** Language process is generally undergoing in the left-hemisphere hemisphere of the brain.

22 The rate of development of one side of the body has influence on hemisphere preference in fetus.

Questions 23-26

Do the following statements agree with the information given in Reading Passage 2?

In boxes 23-26 on your answer sheet write

YES if the statement is true

NO if the statement is false

NOT GIVEN if the information is not given in the passage

23 The study of twins shows that genetic determination is not the only factor for left-handedness.

24 The number of men with left-handedness is more than that of women.

25 Marc Dax's report was widely recognized in his time.

26 John Wada based his findings on his research of people with language problems.

Passage 3

Learning to Walk

These days the feet of a typical city dweller rarely encounter terrain any more uneven than a crack in the pavement. While that may not seem like a problem, it turns out that by flattening our urban environment we have put ourselves at risk of a surprising number of chronic illnesses and disabilities. Fortunately, the commercial market has come to the rescue with a choice of products. Research into the idea that flat floors could be detrimental to our health was pioneered back in the late 1960s in Long Beach, California. Podiatrist Charles Brantingham and physiologist Bruce Beekman were concerned with the growing epidemic of high blood pressure, varicose veins and deep-vein thromboses and reckoned they might be linked to the uniformity of the surfaces that we tend to stand and walk on.

The trouble, they believed, was that walking continuously on flat floors, sidewalks and streets concentrates forces on just a few areas of the foot. As a result, these surfaces are likely to be far more conducive to chronic stress syndromes than natural surfaces, where the foot meets the ground in a wide variety of orientations. They understood that the anatomy of the foot parallels that of the human hand - each having 26 bones, 33 joints and more than 100 muscles, tendons and ligaments - and that modern lifestyles waste all this potential flexibility.

Brantingham and Beekman became convinced that the damage could be rectified by making people wobble. To test their ideas, they got 65 factory workers to try standing on a variable terrain floor - spongy mats with varying degrees of resistance across the surface. This modest irregularity allowed the soles of the volunteers' feet to deviate slightly from the horizontal each time they shifted position. As the researchers hoped, this simple intervention made a huge difference, within a few weeks. Even if people were wobbling slightly, it activated a host of muscles in their legs, which in turn helped pump blood back to their hearts. The muscle action prevented the pooling of blood in their feet and legs, reducing the stress on the heart and circulation. Yet decades later, the flooring of the world's largest workplaces remains relentlessly smooth. Earlier this year, however, the idea was revived when other researchers in the US announced findings from a similar experiment with people over 60. John Fisher and colleagues at the Oregon Research Institute in Eugene designed a mat intended to replicate the effect of walking on cobblestones*.

In tests funded by the National Institute of Aging, they got some 50 adults to walk on the toots in their bare feet for less than an hour, three times a week. After 16 weeks, these people showed marked improvements in mobility, and even a significant reduction in blood pressure. People in a control group who walked on ordinary floors also improved but not as dramatically. The mats are now available for purchase and production is being scaled up. Even so, demand could exceed supply if this foot-stimulating activity really is a 'useful non-pharmacological approach for preventing or controlling hypertension of older adults, as the researchers believe. They are not alone in recognising the benefits of cobblestones. Reflexologists have long advocated walking on textured surfaces to stimulate so-called 'acupoints' on the soles of the feet. They believe that pressure applied to particular spots on the foot connects directly to particular organs of the body and somehow enhances their function. In China, spas, apartment blocks and even factories promote their cobblestone paths as healthful amenities. Fisher admits he got the concept from regular visits to the country. Here, city dwellers take daily walks along cobbled paths for five or ten minutes, perhaps several times a day, to improve their health. The idea is now taking off in Europe too.

People in Germany, Austria and Switzerland can now visit 'barefoot parks' and walk along 'paths of the senses - with mud, logs, stone and moss underfoot. And it is not difficult to construct your own path with simple everyday objects such as stones or bamboo poles. But if none of these solutions appeal, there is another option. A new shoe on the

market claims to transform flat, hard, artificial surfaces into something like uneven ground. 'These shoes have an unbelievable effect,' says Benno Nigg, an exercise scientist at Calgary University in Canada.

Known as the Masai Barefoot Technology, the shoes have rounded soles that cause you to rock slightly when you stand still, exercising the small muscles around the ankle that are responsible for stability. Forces in the joint are reduced, putting less strain on the system, Nigg claims.

Some of these options may not appeal to all consumers and there is a far simpler alternative.

If the urban environment is detrimental to our health, then it is obvious where we should turn. A weekend or even a few hours spent in the countryside could help alleviate a sufferer's aches and pains, and would require only the spending of time.

However, for many modern citizens, the countryside is not as accessible as it once was and is in fact a dwindling resource. Our concrete cities are growing at a terrifying rate - perhaps at the same rate as our health problems.

Questions 27-31

Do the following statements agree with the information given in Reading Passage?

In boxes **27-31** on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

27..... Brantingham and Beekman were the first researchers to investigate the relationship between health problems and flat floors.

28..... The subjects in Fisher's control group experienced a decline in their physical condition.

29..... The manufacturers are increasing the number of cobblestone mats they are making.

30..... Fisher based his ideas on what he saw during an overseas trip.

31..... The Masai Barefoot Technology shoes are made to fit people of all ages.

Questions 32-34

Choose the correct letter, **A, B, C or D**.

Write the correct letter in boxes **32-34** on your answer sheet.

32 The writer suggests that Brantingham and Beekman's findings were

- A** ignored by big companies.
- B** doubted by other researchers.
- C** applicable to a narrow range of people.
- D** surprising to them.

33 What claim is made by the designers of the cobblestone mats"?

- A** They need to be used continuously in order to have a lasting effect.
- B** They would be as beneficial to younger people as to older people.
- C** They could be an effective alternative to medical intervention.
- D** Their effects may vary depending on individual users.

34 Which of the following points does the writer make in the final paragraph?

- A** People should question new theories that scientists put forward.
- B** High prices do not necessarily equate to a quality product.
- C** People are setting up home in the country for health reasons.

D The natural environment is fast disappearing.

Questions 35-40

Complete the summary below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes **35-40** on your answer sheet.

In their research, Brantingham and Beekman looked at the complex physical **35**..... of the foot and noted that the surfaces of modern environments restrict its movement. They invented a mat which they tried out on factory workers. Whenever the workers walked on it, the different levels of **36**..... in the mat would encourage greater muscle action. In turn, this lessened the effect of **37**..... on the cardiovascular system. Similar research was undertaken by John Fisher and colleagues in Oregon. As a result of their findings, they decided to market cobblestone mats to the elderly as a means of dealing with **38**..... Reflexologists claim that by manipulating specific parts of the feet, the performance of certain **39**..... will also improve. Finally, Benno Nigg at Calgary University believes that specially shaped **40**..... on shoes should give health benefits

Mind Music

Scientists investigate 'earworms', the music we can't get out of our head

A Ever had a song stuck in your head, playing on an endless loop? Scientists call them 'involuntary musical images', or 'earworms', and a wave of new research is shining light on why they occur and what can be learned from them. Some neuroscientists and cognitive psychologists are studying earworms to explore the mysteries of memory and the part of the brain that is beyond our conscious control. The idea that we have full control over our thought processes is an illusion,' says psychologist Lauren Stewart, who founded the master's program in music, mind and brain at Goldsmiths, University of London, UK, where recent research has taken place. Researchers haven't been able to watch what happens in the brain when earworms occur, because they happen unpredictably. Much of what is known about them comes from surveys, questionnaires, diaries and lab experiments.

B A Goldsmith study published in the journal Memory and Cognition this year showed that the singing we hear in our heads tends to be true to actual recordings. Researchers had 17 volunteers tap to the beat of any earworm they heard during a four-day period while a device attached to their wrist recorded their movements. The tapping tempos were within 10% of the tempos of the original recordings. Another Goldsmiths study, published this year in Consciousness and Cognition, found that people who report hearing earworms often, and find them most intrusive, have slightly different brain structures, with more gray matter in areas associated with processing emotions.

C Studies also show that the music in our heads often starts playing during times of 'low cognitive load', such as while showering, getting dressed, walking, or doing chores. Dr Stewart likens earworms to 'sonic screen savers' that keep the mind entertained while it is otherwise unoccupied. She and her colleagues tested that theory by having volunteers listen to songs and giving them various tasks afterward. The volunteers who sat idly for the next five minutes were the most likely to report hearing the music in their heads. Dr Stewart observed that the more challenging the activity, the less likely the volunteers were to hear the music. Diary studies also show songs tend to match people's moods and therefore they are not random. If you are energized and upbeat, an earworm that occurs is likely to be uptempo too.

D Songs the brain fixates on are usually those it has been exposed to recently, surveys show, which is why tunes getting heavy radio play frequently top the earworm charts. Even tunes you may have heard but didn't pay attention to can worm their way into your subconscious, says Ira Hyman, a psychologist at Western Washington University in

Bellingham, USA. In an unpublished study there, participants who listened to music while doing other tasks were more likely to report that the songs returned as earworms later on, compared with participants who simply listened.

E Some earworms are just fragments of a song that repeat like a broken record. So, when the mind hits a part of a song it can't remember, it loops back rather than moving on. That could make an earworm even more entrenched, Dr Hyman says. According to a theory known as the Zeigarnik effect, named for a Soviet psychologist, Bluma Zeigarnik, unfinished thoughts and activities weigh on the mind more heavily than those that are completed, although experiments exposing students to interrupted songs have yielded mixed results.

F Researchers say they can't pinpoint a spot in the brain where earworms live. Imaging studies by Andrea Halpern at Bucknell University, in Lewisburg, USA, have shown that deliberately imagining music and actually listening to music activate many of the same neurological networks. Dr Halpern's earlier studies showed that when subjects listened to the first few notes of familiar music, areas in the right frontal and superior temporal portions of the brain became activated, along with the supplementary motor area at the top, which is typically involved in remembering sequences. When the same subjects listened to unfamiliar music and were asked to recall it, there was activity in the left frontal portions of the brain instead.

G One factor that makes some songs stick might be repetition. 'Repetition leads to familiarity which leads to anticipation, which is satisfied by hearing the song,' says John Seabrook, author of *The Song Machine: Inside the Hit Factory*, about how producers pump pop songs full of aural 'hooks', the punchy melodic phrases designed to target the brain and leave it wanting more. The researchers are comparing the melodic structure of 100 often-mentioned songs with 100 similarly popular songs that weren't cited as earworms, to assess the difference. Songs with earworm potential appear to share certain features: a repeating pattern of ups and downs in pitch, and an irregular musical interval.

H The researchers plan next to test their results in reverse, and play ringtones from songs of both the earworm and non-earworm variety for volunteers several times a day to see which ones get stuck. Drs Stewart and Halpern are now working together to recruit survey participants for a study looking at whether people at different stages of life experience earworms differently. 'You can argue that older people might get them more often because they know more songs,' Dr Halpern says. 'But the few responses we have so far indicate that they have earworms less often. It could be that they don't play music as often as younger people do.'

Questions 28-31

The reading Passage has eight paragraphs, **A-H**. Which paragraph contains the following information?

Write the correct letter, **A-H**, in boxes **28-31** on your answer sheet.

28..... a description of the characteristics common to songs with earworms

29..... a justification for research into earworms

30..... a description of the brain's reaction to known and unknown songs

31..... details of proposed research into the frequency with which earworms occur in different age groups

Questions 32-35

Complete the summary below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes **32-35** on your answer sheet.

Goldsmiths study

Researchers from Goldsmiths concluded that the music we imagine in our minds is quite similar to recordings. They

proved this by asking volunteers to record the rhythm of music using a monitor on their **32**..... Further research has demonstrated that those who hear earworms more frequently have brains that may deal with **33**..... differently from other people, Dr Stewart also believes that the brain is **34**..... by earworms when it is not focused on a task. In fact, a reduction in the occurrence of earworms was found to be directly related to how **35**..... the task was Interestingly, volunteers' diaries revealed that the songs they heard inside their head reflected their moods, so the choice of music is not accidental.

Questions 36-40

Look at the following statements and the list of researchers below.

Match each statement with the correct person, **A, B, C or D**.

Write the correct letter. A, B, C or D, in boxes **36-40** on your answer sheet.

NB You may use any letter more than once.

36..... Some musicians create music that is intentionally memorable.

37..... People are unable to completely regulate how they think.

38..... We can remember songs without knowing that we have heard them.

39..... Thinking about music has a similar effect on the brain to hearing music.

40..... Earworms are more persistent when only a short section of the song is constantly replayed.

List of Researchers

A Lauren Stewart

B Ira Hyman

C Andrea Haiper

D John Seabrook

The role of accidents in business

In 1894 Dr John Kellogg and his brother. Will, was supervising a hospital and health spa in Michigan. The patients were on a restricted diet. One day, the brothers left cooked wheat untended for more than 24 hours. When they returned, they saw what they had done. It was no good to eat, but they decided to run the stale wheat through rollers, just to see how it would turn out. Normally, the process produced long sheets, but they were surprised to discover that this time the rollers created flat flakes. They baked them and then tried the same thing with corn. From this accidental discovery came the cornflakes that generations have now been eating for breakfast.

Accidents happen; there is nothing predictable and orderly about innovation. Nobel laureate Sir Alan Hodgkin, who discovered how nerve cells transmit electrical impulses between the skin and the brain, commented: 'I believe that the record of my published papers conveys an impression of directedness and planning which does not at all coincide with the actual sequence of events.'

The same rule applies in business. The mistake that gave US cornflakes keeps repeating itself in the history of disruptive innovation, the kind that transforms markets. Louis Daguerre, for, instance, discovered the technique that gave US photography in the 1830s, when drops of mercury from a shattered thermometer produced a photographic image. The microwave was discovered when Peroy Spender, a scientist with Raytheon, was testing a new vacuum tube and discovered that the sweet in his pocket had melted. The artificial sweetener, saccharin, was the unintentional result of a medical scientist's work on a chemical treatment for gastric ulcers. While working for the firm 3M, researcher Art Fry had no idea he was taking the first steps towards Post-It Notes when he used bits of adhesive office paper that could be easily lifted off the page to replace the scrap paper bookmarks that kept falling out of his hymn book.

Breakthrough and disruptive innovation are rarely driven by orderly processes. Usually, they come out of a chaotic, haphazard mess, which is why big companies, full of managers schooled in business programs designed to eliminate random variation and mistakes, struggle with them. In these sorts of environments, accidents are called failures and are discouraged.

It is no surprise then that research from the late British economist Paul Geroski and London Business School's Constantinos Markides found that companies that were skilled at innovation were usually not that skilled when it came to commercialization, and vice versa. Their book, *Fast Second*, divides businesses into 'colonists' and 'consolidators'. Small and nimble, colonists are adept at creating market niches but are terrible institution builders. Consolidators, with their strong cultures of discipline and cost control, know how to take clever ideas from other firms and turn them into mass market items. Microsoft is a prime instance of this.

With companies spending hundreds of billions of dollars on research and development, US academics Robert Austin and Lee Devin examined how managers can encourage productive slip-ups. In their article *Accident, Intention and Expectation in the Innovation Process*, they argue that business processes actually prevent helpful missteps from occurring. According to their catalogue of accidents, not all false steps and mishaps are equal. Accidents, they say, come from unlikely mental associations such as memories and vague connections, looking for something and finding it in an unexpected way, looking for one thing and finding something else, and not looking for anything but finding something valuable.

Accident-prone innovation, they say, requires companies to get outside the 'cone of expectation'. It means throwing together groups from diverse backgrounds and combining ideas in unpredictable ways, other strategies also include having systems that watch out for accidents and examine them for value, generating them when they do not happen often enough, seizing only the useful ones, capturing their valuable features, and building on them to add value and give potential for useful accidents.

All this, however, requires thinking that is often counter-intuitive to the way businesses operate. In other words, it is the kind of thinking that goes against the beliefs of most business managers. It runs counter to the notion frequently pushed by consultants that you can 'harness' creativity and direct it to line up with intention. 'The cost of accidents business, people tend to call such efforts failure.'

There are tentative signs that more companies are starting to realise that failure can lead to commercial gain, and that this is part of the risk-taking that underpins innovation. Australia's largest brewing company, for example, made a bad error when it launched a new beer called Empire Lager, pitched at younger consumers. Having spent a fortune creating a beer with a sweeter taste, designing a great-looking bottle and a television campaign, Foster's was left with a drink that no one wanted to buy. The target market was more interested in brands built up by word of mouth. Instead of wiping the unsuccessful product launch, Fosters used this lesson learned to go on and develop other brands instead. One of them, Pure Blonde, is now ranked as Australia's fifth-largest beer brand. Unlike Empire Lager, there has been almost no promotion and its sales are generated more by word of mouth.

Other companies are taking similar steps to study their own slip-ups. Intuit, the company behind financial tools such as Quicken, holds regular 'When Learning Hurts' sessions. But this sort of transformation is never easy. In a market that focuses on the short-term, convincing employees and shareholders to tolerate failure and not play it safe is a

big thing to ask.

Questions 27-31

Do the following statements agree with the claims of the writer in Reading Passage?

In boxes **27-31** on your answer sheet, write

YES if the statement agrees with the claims of the writer

NO if the statement contradicts the claims of the writer

NOT GIVEN if it is impossible to say what the writer thinks about this

27..... The delay in the process used by the Kellogg brothers affected the final product.

28..... Sir Alan Hodgkin is an example of someone whose work proceeded in a logical and systematic way.

29..... Daguerre is an exception to the general rule of innovation.

30..... The discovery of saccharin occurred by accident during drug research.

31..... The company 3M should have supported Art Fry by funding his idea of PostIt Notes.

Questions 32-35

Complete each sentence with the correct ending, **A-H**, below.

Write the correct letter, **A-H**, in boxes **32-35** on your answer sheet.

32..... The usual business environment

33..... Geroki and Markides's book

34..... Microsoft is an example of a company which

35..... The origin of useful accidents

A can be found in unusual thoughts and chance events.

B can be taught in business schools.

c has made a success from someone else's invention.

D is designed to nurture differences.

E is unlikely to lead to creative innovation

F says that all mistakes are the same.

G shows that businesses are good at either inventing or selling.

H suggests ways of increasing the number of mistakes

Questions 36-40

Choose the correct letter, **A, B, c or D**

Write the correct letter in boxes **36-40** on your answer sheet.

36 How do Austin and Devin advise companies to get out of the 'cone of expectation'?

A by decreasing the number of company systems

B by forming teams of different types of people

C by hiring new and creative people

D by holding regular brainstorming meetings

37 In recommending 'counter-intuitive' thinking, what do Austin and Devin imply?

A that failing at business is bad for staff morale

B that innovation cannot be planned for

C that most businesses should be devoted to avoiding mistakes

D that the cost of mistakes is an important consideration

38 The writer describes the Empire Lager disaster in order to show that

A success can come out of a business failure

B the majority of companies now value risk-taking.

C TV advertising works better on older people

D young beer drinkers do not like a sweet taste

39 Pure Blonde has been more successful than Empire Lager because

A digital media other than TV were used.

B it was advertised under a different brand name.

C it was launched with very little advertising.

D the advertising budget was larger

40 The writer concludes that creating a culture that learns from mistakes

A brings short-term financial gains.

B can be very difficult for some companies.

C holds no risk for workers.

D is a popular move with shareholders.

Global Warming in New Zealand

For many environmentalists, the world seems to be getting warmer. As the nearest country of South Polar Region, New Zealand has maintained an upward trend in its average temperature in the past few years. However, the

temperature in New Zealand will go up 4°C in the next century while the polar region will go up more than 6°C.

The different pictures of temperature stem from its surrounding ocean which acts like the air conditioner. Thus New Zealand is comparatively fortunate.

Scientifically speaking, this temperature phenomenon in New Zealand originated from what researchers call "SAM" (Southern Annular Mode), which refers to the wind belt that circles the Southern Oceans including New Zealand and Antarctica. Yet recent work has revealed that changes in SAM in New Zealand have resulted in a weakening of moisture during the summer, and more rainfall in other seasons. A bigger problem may turn out to be heavier droughts for agricultural activities because of more water loss from soil, resulting in poorer harvest before winter when the rainfall arrive too late to rescue.

Among all the calamities posed by drought, moisture deficit ranks the first, Moisture deficit is the gap between the water plants need during the growing season and the water the earth can offer. Measures of moisture deficit were at their highest since the 1970s in New Zealand. Meanwhile, ecological analyses clearly show moisture deficit is imposed at different growth stage of crops. If moisture deficit occurs around a crucial growth stage, it will cause about 22% reduction in grain yield as opposed to moisture deficit at vegetative phase.

Global warming is not only affecting agriculture production. When scientists say the country's snow pack and glaciers are melting at an alarming rate due to global warming, the climate is putting another strain on the local places. For example, when the development of global warming is accompanied by the falling snow line, the local skiing industry comes into a crisis. The snow line may move up as the temperature goes up, and then the snow at the bottom wilt melt earlier. Fortunately, it is going to be favourable for the local skiing industry to tide over tough periods since the quantities of snowfall in some areas are more likely to increase. What is the reaction of glacier region? The climate change can be reflected in the glacier region in southern New Zealand or land covered by ice and snow. The reaction of a glacier to a climatic change involves a complex chain of processes. Over time periods of years to several decades, cumulative changes in mass balance cause volume and thickness changes, which will affect the flow of ice via altered internal deformation and basal sliding. This dynamic reaction finally leads to glacier length changes, the advance or retreat of glacier tongues. Undoubtedly, glacier mass balance is a more direct signal

of annual atmospheric conditions. The latest research result of National Institute of Water and Atmospheric (NIWA) Research shows that glaciers line keeps moving up because of the impacts of global warming. Further losses of ice can be reflected in Mr. Cook Region. By 1996, a 14 km long sector of the glacier had melted down forming a melt take (Hooker Lake) with a volume. Melting of the glacier front at a rate of 40 m/yr will cause the glacier to retreat at a rather uniform rate. Therefore, the lake will continue to grow until it reaches the glacier bed.

A direct result of the melting glaciers is the change of high tides that serves the main factor for sea level rise. The trend of sea level rise will bring a threat to the groundwater system for its hyper-saline groundwater and then pose a possibility to decrease the agricultural production. Many experts believe that the best way to counter this trend is to give a longer-term view of sea level change in New Zealand. Indeed, the coastal boundaries need to be upgraded and redefined. There is no doubt that global warming has affected New Zealand in many aspects. The emphasis on the global warming should be based on the joints efforts of local people and experts who conquer the tough period. For instance, farmers are taking a long term, multi-generational approach to adjust the breeds and species according to the temperature. Agriculturists also find ways to tackle the problems that may bring to the soil. In broad terms, going forward, the systemic resilience that's been going on a long time in the ecosystem will continue.

How about animals' reaction? Experts have surprisingly realised that animals have unconventional adaptations to global warming. A study has looked at sea turtles on a few northern beaches in New Zealand and it is very interesting to find that sea turtles can become male or female according to the temperature. Further researches will try to find out how rising temperatures would affect the ratio of sex reversal in their growth. Clearly, the temperature of the nest plays a vital role in the sexes of the baby turtles.

Tackling the problems of global warming is never easy in New Zealand, because records show the slow process of global warming may have a different impact on various regions. For New Zealand, the emission of carbon dioxide only accounts for 0.5% of the world's total, which has met the governmental standard. However, New Zealand's effort counts only a tip of the iceberg. So far, global warming has been a world issue that still hangs in an ambiguous future.

Questions 27-32

Choose the correct letter, A, B, C or D.

Write the correct letter in boxes **27-32** on your answer sheet.

27 What is the main idea of the first paragraph?

- A The temperature in the polar region will increase less than that in New Zealand in the next century.
- B The weather and climate of New Zealand is very important to its people because of its close location to the polar region.
- C The air condition in New Zealand will maintain a high quality because of the ocean.
- D The temperature of New Zealand will increase less than that of other regions in the next 100 years because it is surrounded by sea.

28 What is one effect of the wind belt that circles the Southern Oceans?

- A New Zealand will have more moisture in winds in summer.
- B New Zealand needs to face droughts more often in hotter months in a year.
- C Soil water will increase as a result of weakening moisture in the winds.
- D Agricultural production will be reduced as a result of more rainfall in other seasons.

29 What does "moisture deficit" mean to the grain and crops?

- A The growing condition will be very tough for crops.
- B The growing season of some plants can hardly be determined.
- C There will be a huge gap between the water plants needed and the water the earth can offer.
- D The soil of grain and crops in New Zealand reached its lowest production since 1970s.

30 What changes will happen to skiing industry due to the global warming phenomenon?

- A The skiing station may lower the altitude of skiing.
- B Part of the skiing station needs to move to the north.
- C The snowfall may increase in part of the skiing station.
- D The local skiing station may likely to make a profit because of the snowfall increase.

31 Cumulative changes over a long period of time in mass balance will lead to

- A alterations in the volume and thickness of glaciers.
- B faster changes in internal deformation and basal sliding.
- C larger length of glaciers.
- D retreat of glacier tongues as a result of change in annual atmospheric conditions.

32 Why does the writer mention NIWA in the sixth paragraph?

- A To use a particular example to explain the effects brought by glacier melting.
- B To emphasize the severance of the further loss of ice in Mt. Cook Region.
- C To alarm the reader of melting speed of glaciers at a uniform rate.
- D To note the lake in the region will disappear when it reach the glacier bed.

Questions 33-35

Complete the summary below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Write your answers in boxes **33-35** on your answer sheet.

Research data shows that sea level has a closely relation with the change of climate. The major reason for the increase in sea level is connected with **33** _____. The increase in sea level is also said to have a threat to the underground water system, the destruction of which caused by rise of sea level will lead to a high probability of reduction in **34** _____. In the long run, New Zealanders may have to improve the **35** _____ if they want to diminish the effect change in sea levels.

Questions 36-40

Do the following statements agree with the claims of the writer in Reading Passage 3?

In boxes **36-40** on your answer sheet, write

- YES if the statement agrees with the claims of the writer
- NO if the statement contradicts the claims of the writer
- NOT GIVEN if it is impossible to say what the writer thinks about this

36 Farmers are less responsive to climate change than agriculturists.

37 Agricultural sector is too conservative and resistant to deal with climate change.

38 Turtle is vulnerable to climate change.

39 The global warming is going slowly, and it may have different effects on different areas in New Zealand.

40 New Zealand must cut carbon dioxide emissions if they want to solve the problem of global warming.

The Significant Role of Mother Tongue in Education

One consequence of population mobility is an increasing diversity within schools. To illustrate, in the city of Toronto in Canada, 58% of kindergarten pupils come from homes where English is not the usual language of communication. Schools in Europe and North America have experienced this diversity for years, and educational policies and

practices very widely between countries and even within countries. Some political parties and groups search for ways to solve the problem of diverse communities and their integration in schools and society. However, they see few positive consequences for the host society and worry that this diversity threatens the identity of the host society. Consequently, they promote unfortunate educational policies that will make the "problem" disappear. If students retain their culture and language, they are viewed as less capable of identifying with the mainstream culture and learning the mainstream language of the society.

The challenge for educators and policy-makers is to shape the evolution of national identity in such a way that the rights of all citizens (including school children) are respected, and the cultural, linguistic, and economic resources of the nation are maximised. To waste the resources of the nation by discouraging children from developing their mother tongues is quite simply unintelligent from the point of view of national self-interest. A first step in providing an appropriate education for culturally and linguistically diverse children is to examine what the existing research says about the role of children's mother tongues in their educational development.

In fact, the research is very clear. When children continue to develop their abilities in two or more languages throughout their primary school, they gain a deeper understanding of language and how to use it effectively. They have more practice in processing language, especially when they develop literacy in both. More than 150 research studies conducted during the past 35 years strongly support what Goethe, the famous eighteenth-century German philosopher, once said: the person who knows only one language does not truly know that language. Research suggests that bilingual children may also develop more flexibility in their thinking as a result of processing information through two different languages.

The level of development of children's mother tongue is a strong predictor of their second language development. Children who come to school with a solid foundation in their mother tongue develop stronger literacy abilities in the school language. When parents and other caregivers (e.g. grandparents) are able to spend time with their children and tell stories or discuss issues with them in a way that develops their mother tongue, children come to school well-prepared to learn the school language and succeed educationally. Children's knowledge and skills transfer across languages from the mother tongue to the school language. Transfer across languages can be two-way: both languages nurture each other when the educational environment permits children access to both languages.

Some educators and parents are suspicious of mother tongue-based teaching programs because they worry that they take time away from the majority language. For example, in a bilingual program where 50% of the time is spent teaching through children's home language and 50% through the majority language, surely children won't progress as far in the latter? One of the most strongly established findings of educational research, however, is that well-implemented bilingual programs can promote literacy and subject-matter knowledge in a minority language without any negative effects on children's development in the majority language. Within Europe, the Foyer program in Belgium, which develops children's speaking and literacy abilities in three languages (their mother tongue, Dutch and French), most clearly illustrates the benefits of bilingual and trilingual education (see Cummins, 2000).

It is easy to understand how this happens, when children are learning through a minority language, they are learning concepts and intellectual skill too. Pupils who know how to tell the time in their mother tongue understand the concept of telling time. In order to tell time in the majority language, they do not need to re-learn the concept.

Similarly, at more advanced stages, there is transfer across languages in other skills such as knowing how to distinguish the main idea from the supporting details of a written passage or story, and distinguishing fact from opinion. Studies of secondary school pupils are providing interesting findings in this area, and it would be worth extending this research.

Many people marvel at how quickly bilingual children seem to "pick up" conversational skills in the majority language at school (although it takes much longer for them to catch up with native speakers in academic language skills). However, educators are often much less aware of how quickly children can lose their ability to use their mother tongue, even in the home context. The extent and rapidity of language loss will vary according to the concentration of families from a particular linguistic group in the neighborhood. Where the mother tongue is used extensively in the community, then language loss among young children will be less. However, where language communities are not concentrated in particular neighborhoods, children can lose their ability to communicate in their mother tongue within 2-3 years of starting school. They may retain receptive skills in the language but they will use the majority language in speaking with their peers and siblings and in responding to their parents. By the time children become adolescents, the linguistic division between parents and children has become an emotional chasm. Pupils frequently become alienated from the cultures of both home and school with predictable results.

Questions 27-30

Choose the correct letter, A, B, C or D.

Write the correct letter in boxes **27-30** on your answer sheet.

27 What point did the writer make in the second paragraph?

- A Some present studies on children's mother tongues are misleading.
- B A culturally rich education programme benefits some children more than others.
- C Bilingual children can make a valuable contribution to the wealth of a country.
- D The law on mother tongue use at school should be strengthened.

28 Why does the writer refer to something that Goethe said?

- A to lend weight to his argument
- B to contradict some research
- C to introduce a new concept
- D to update current thinking

29 The writer believes that when young children have a firm grasp of their mother tongue

- A They can teach older family members what they learnt at school.
- B They go on to do much better throughout their time at school.
- C They can read stories about their cultural background.
- D They develop stronger relationships with their family than with their peers.

30 Why are some people suspicious about mother tongue-based teaching programmes?

- A They worry that children will be slow to learn to read in either language.
- B They think that children will confuse words in the two languages.
- C They believe that the programmes will make children less interested in their lessons.
- D They fear that the programmes will use up valuable time in the school day.

Questions 31-35

Complete the summary using the list of words, **A-J**, below.

Write the correct letter, **A-J**, in boxes **31-35** on your answer sheet.

Bilingual Children

It was often recorded that bilingual children acquire the 31 _____ to converse in the majority

language remarkable quickly. The fact that the mother tongue can disappear at a similar **32** _____ is less well understood. This phenomenon depends, to a certain extent, on the proposition of people with the same linguistic background that have settled in a particular **33** _____. If this is limited, children are likely to lose the active use of their mother tongue. And thus no longer employ it even with **34** _____, although they may still understand it. It follows that teenager children in these circumstances experience a sense of **35** _____ in relation to all aspects of their lives.

- | | | |
|-----------------|----------|---------------|
| A teachers | B school | C dislocation |
| D rate | E time | F family |
| G communication | H type | I ability |
| J area | | |

Questions 36-40

Do the following statements agree with the views of the writer in Reading passage 3?

In boxes **36-40** on your answer sheet, write

YES if the statement agrees with the views of the writer

NO if the statement contradicts the views of the writer

NOT GIVEN if it is impossible to say what the writer thinks about this

36 Less than half of the children who attend kindergarten in Toronto have English as their mother tongue.

37 Research proves that learning the host country language at school can have an adverse effect on a child's mother tongue.

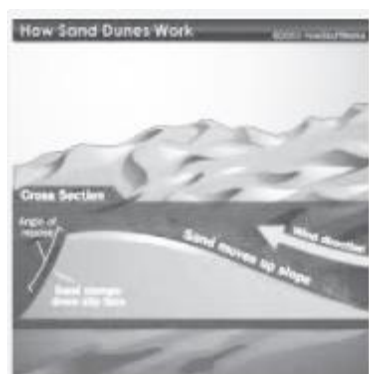
38 The Foyer program is accepted by the French education system.

39 Bilingual children are taught to tell the time earlier than monolingual children.

40 Bilingual children can apply reading comprehension strategies acquired in one language when reading in the other.

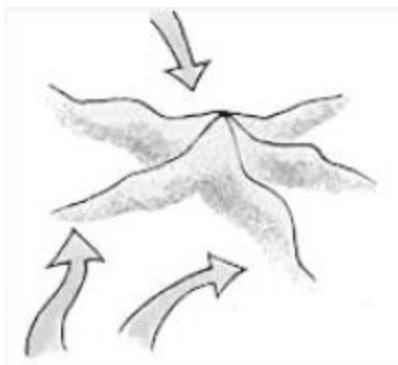
Sand Dunes

A One of the main problems posed by sand dunes is their encroachment on human habitats. Sand dunes move by different means, all of them aided by the wind. Sand dunes threaten buildings and crops in Africa, the Middle East, and China. Preventing sand dunes from overwhelming cities and agricultural areas has become a priority for the United Nations Environment Program. On the other hand, dune habitats provide niches for highly specialized plants and animals, including numerous rare and endangered species.



B Sand is usually composed of hard minerals such as quartz that cannot be broken down into silt or clay. Yellow, brown and reddish shades of sand indicate the presence of iron compounds. Red sand is composed of quartz coated

by a layer of iron oxide. White sands are nearly pure gypsum. Sand with a high percentage of silicate can be used in glassmaking. Sandstone is created by sand, mixed with lime, chalk or some other material that acts as a binding agent that is deposited in layers at the bottom of a sea or other areas and pressed together into rock by the great pressure of sediments that are deposited on top of it over thousands or millions of years.



C The most common dune form on Earth and on Mars is the crescentic. Crescent-shaped mounds are generally wider than they are long. The slip faces are on the concave sides of the dunes. These dunes form under winds that blow consistently from one direction, and they also are known as barchans, or transverse dunes. Some types of crescentic dunes move more quickly over desert surfaces than any other type of dune. A group of dunes moved more than 100 metres per year between 1954 and 1959 in the China's Ningxia Province, and similar speeds have been recorded in the Western Desert of Egypt. The largest crescentic dunes on Earth, with mean crest-to-crest widths of more than 3 kilometres, are in China's Taldamakan Desert.

D Radially symmetrical, star dunes are pyramidal sand mounds with slip faces on three or more arms that radiate from the high center of the mound. They tend to accumulate in areas with multidirectional wind regimes. Star dunes grow upward rather than laterally. They dominate the Grand Erg Oriental of the Sahara. In other deserts, they occur around the margins of the sand seas, particularly near topographic barriers. In the southeast Badain Jaran Desert of China, the star dunes are up to 500 metres tall and may be the tallest dunes on Earth. Straight or slightly sinuous sand ridges typically much longer than they are wide are known as linear dunes. They may be more than 160 kilometres (99 mi) long. Some linear dunes merge to form Y-shaped compound dunes. Many form in bidirectional wind regimes. The long axes of these dunes extend in the resultant direction of sand movement. Linear loess hills known as pahas are superficially similar.

E Once sand begins to pile up, ripples and dunes can form. Wind continues to move sand up to the top of the pile until the pile is so steep that it collapses under its own weight. The collapsing sand comes to rest when it reaches just the right steepness to keep the dune stable. This angle, usually about 30-34°, is called the angle of repose. Every pile of loose particles has a unique angle of repose, depending upon the properties of the material it's made of, such as the grain size and roundness. Ripples grow into dunes with increase of wind and sand input.

F The repeating cycle of sand inching up the windward side to the dune crest, then slipping down the dune's slip face allows the dune to inch forward, migrating in the direction the wind blows. As you might guess, all of this climbing then slipping leaves its mark on the internal structure of the dune. The image on the right shows fossil sand dune structure preserved in the Merced Formation at Fort Funston, Golden Gate National Recreation Area. The sloping lines or laminations you see are the preserved slip faces of a migrating sand dune. This structure is called cross-bedding, and can be the result of either wind or water currents. The larger the cross-bedded structure, however, the more likely it is to be formed by wind, rather than water.

G Sand dunes can 'sing' at a level up to 115 decibels and generate sounds in different notes. The dunes at Sand Mountain in Nevada usually sing in a low C but can also sing in B and C sharp. The La Mar de Dunas in Chile hum in F while those at the Ghord Lahmar in Morocco howl in G sharp. The sounds are produced by avalanches of sand generated by blowing winds. For a while it was thought that the avalanches caused the entire dune to resonate like a flute or violin but if that were true then different size dunes would produce different notes. In the mid 2000s, American, French and Moroccan scientists visiting sand dunes in Morocco, Chile, China and Oman published a paper in the Physical Review Letters that determined the sounds produced by collisions between grains of sand that caused the motions of the grains to become synchronized, causing the outer layer of a dune to vibrate like the cone of a loudspeaker, producing sound. The tone of the sounds depended primarily on the size of the grains.

H Scientists performed a computer simulation on patterns and dynamics of desert dunes in laboratory. Dune patterns observed in deserts were reproduced. From the initial random state, stars and linear dunes are produced, depending on the variability of the wind direction. The efficiency in sand transport is calculated through the course of development. Scientists found that the sand transport is the most efficient in the linear transverse dune. The efficiency in sand transport always increased through the evolution, and the way it increase was stepwise. They also found that the shadow zone, the region where the sand wastes the chance to move, shrinks through the course of evolution, which greatly helps them build a model to simulate sand move.

Questions 27-34

Reading Passage 3 has eight paragraphs, **A-H**.

Choose the correct heading for paragraphs **A-H** from the list below.

Write the correct number, **i-x**, in boxes **1-8** on your answer sheet.

List of Headings

- i potential threat to buildings and crops despite of benefit.
- ii the cycle of sand moving forward with wind
- iii protection method in various countries.
- iv scientists simulate sand move and build model in lab
- v sand composition explanation
- vi singing sand dunes
- vii other types of sand dunes
- viii the personal opinion on related issues.
- ix reasons why sand dunes form
- x the most common sand type

27 Paragraph A

28 Paragraph B

29 Paragraph C

30 Paragraph D

31 Paragraph E

32 Paragraph F

33 Paragraph G

34 Paragraph H

Question 35-36

Answer the questions 35-36 and choose correct letter **A,B,C** or **D**.

35 What is main composition of white sand made of according to the passage?

A Quartz

- B Gypsum
- C Lime
- D Iron

36 Which one is not mentioned as a sand type in this passage?

- A Linear
- B Crescentic
- C Overlap
- D Star

Questions 37-40

Complete the summary using the list of words, **A-J** below.

Write the correct letter, **A-J** in boxes **37-40** on your answer sheet.

Crescentic is an ordinary 37 on both Earth and Mars, apart from which, there are also other types of sand dunes. Different color of the sand reflects different components, some of them are rich in 38 that can not be easily broken into clay. Sand dunes can 'sing' at a level up to 115 decibels and generate sounds in different notes. Sand dunes can be able to 39 at certain level of sound intensity, and different size of grains creates different 40 of the sounds.

- | | | | | |
|------------|------------|------------|--------|--------------|
| A quartz | B shape | C pressure | D tone | E protection |
| F category | G minerals | H sing | I lab | J direction |

Save Endangered Language

'Obviously, we must do some serious rethinking of our priorities, lest linguistics go down in history as the only science that presided obviously over the disappearance of 90 percent of the very field to which it is dedicated.' Michael Krauss, 'The World's Languages in Crisis '.

A Ten years ago Michael Krauss sent a shudder through the discipline of linguistics with his prediction that half the 6, 000 or so languages spoken in the world would cease to be uttered within a century. Unless scientists and community leaders directed a worldwide effort to stabilize the decline of local languages, he warned, nine-tenths of the linguistic diversity of humankind would probably be doomed to extinction. Krauss's prediction was little more than an educated guess, but other respected linguists had been clanging out similar alarms. Kenneth G. Hale of the Massachusetts Institute of Technology noted in the same journal issue that eight languages on which he had done fieldwork had since passed into extinction. A 1990 survey in Australia found that 70 of the 90 surviving aboriginal languages were no longer used regularly by all age groups. The same was true for all but 20 of the 175 Native American languages spoken or remembered in the US. , Krauss told a congressional panel in 1992.



B Many experts in the field mourn the loss of rare languages, for several reasons. To start, there is scientific self-interest: some of the most basic questions in linguistics have to do with the limits of human speech, which are far

from fully explored. Many researchers would like to know which structural elements of grammar and vocabulary—if any—are truly universal and probably therefore hardwired into the human brain. Other scientists try to reconstruct ancient migration patterns by comparing borrowed words that appear in otherwise unrelated languages. In each of these cases, the wider the portfolio of languages you study, the more likely you are to get the right answers.



C Despite the near constant buzz in linguistics about endangered languages over the past 10 years, the field has accomplished depressingly little. ‘You would think that there would be some organized response to this dire situation,’ some attempt to determine which language can be saved and which should be documented before they disappear, says Sarah G. Thomason, a linguist at the University of Michigan at Ann Arbor. ‘But there isn’t any such effort organized in the profession. It is only recently that it has become fashionable enough to work on endangered languages.’ Six years ago, recalls Douglas H. Whalen of Yale University, ‘when I asked linguists who was raising money to deal with these problems, I mostly got blank stares.’ So Whalen and a few other linguists founded the Endangered Languages Fund. In the five years to 2001 they were able to collect only \$80, 000 for research grants. A similar foundation in England, directed by Nicholas Ostler, has raised just \$8, 000 since 1995.

D But there are encouraging signs that the field has turned a corner. The Volkswagen Foundation, a German charity, just issued its second round of grants totaling more than \$2 million. It has created a multimedia archive at the Max Planck Institute for Psycholinguistics in the Netherlands that can house recordings, grammars, dictionaries and other data on endangered languages. To fill the archive, the foundation has dispatched field linguists to document Aweti (100 or so speakers in Brazil), Ega (about 300 speakers in Ivory Coast), Waima’a (a few hundred speakers in East Timor), and a dozen or so other languages unlikely to survive the century. The Ford Foundation has also edged into the arena, its contributions helped to reinvigorate a master-apprentice program created in 1992 by Leanne Hinton of Berkeley and Native Americans worried about the imminent demise of about 50 indigenous languages in California. Fluent speakers receive \$3, 000 to teach a younger relative (who is also paid) their native tongue through 360 hours of shared activities, spread over six months. So far about 5 teams have completed the program, Hinton says, transmitting at least stone knowledge of 25 languages. ‘It’s too early to call this language revitalization,’ Hinton admits. ‘In California, the death rate of elderly speakers will always be greater than the recruitment rate of young speakers. But at least we prolong the survival of the language.’ That will give linguists more time to record these tongues before they vanish.



E But the master-apprentice approach hasn’t caught on outside the U. S., and Hinton’s effort is a drop in the sea. At least 440 languages have been reduced to a mere handful of elders, according to the Ethnologue, a catalogue of languages produced by the Dallas-based group SIL International that crones closest to global coverage. For the vast majority of these languages, there is little or no record of their grammar, vocabulary, pronunciation or use in daily life. Even if a language has been fully documented, all that remains once it vanishes from active use is a fossil skeleton, a scattering of features that the scientist was lucky and astute enough to capture. Linguists may be able to sketch an outline of the forgotten language and fix its place on the evolutionary tree, but little more. ‘How did people start conversations and talk to babies? How did husbands and wives converse?’ Hinton asks. ‘Those are the first things you want to learn when you want to revitalize the language.’

F But there is as yet no discipline of ‘conservation linguistics’, as there is for biology. Almost every strategy tried so far has succeeded in some places but failed in others, and there seems to be no way to predict with certainty what will work where. Twenty years ago in New Zealand, Maori speakers set up ‘language nests’, in which preschoolers were immersed in the native language. Additional Maori-only classes were added as the children progressed through elementary and secondary school. A similar approach was tried in Hawaii, with some success—the number of native speakers has stabilized at 1, 000 or so, reports Joseph E. Grimes of SIL International, who is working on Oahu. Students can now get instruction in Hawaiian all the way through university.



G One factor that always seems to occur in the demise of a language is that the speakers begin to have collective doubts about the usefulness of language loyalty. Once they start regarding their own language as inferior to the majority language, people stop using it for all situations. Kids pick up on the attitude and prefer the dominant language. In many cases, people don’t notice until they suddenly realize that their kids never speak the language, even at home. This is how Cornish and some dialects of Scottish Gaelic is still only rarely used for daily home life in Ireland, 80 years after the republic was founded with Irish as its first official language.

H Linguists agree that ultimately, the answer to the problem of language extinction is multilingualism. Even uneducated people can learn several languages, as long as they start as children. Indeed, most people in the world speak more than one tongue, and in places such as Cameroon (279 languages), Papua New Guinea (823) and India (387) it is common to speak three or four distinct languages and a dialect or two as well. Most Americans and Canadians, to the west of Quebec, have a gut reaction that anyone speaking another language in front of them is committing an immoral act. You get the same reaction in Australia and Russia. It is no coincidence that these are the areas where languages are disappearing the fastest. The first step in saving dying languages is to persuade the world’s majorities to allow the minorities among them to speak with their own voices.

Questions 27-33

The Reading Passage has eight paragraphs, **A-H**

Choose the correct heading for paragraphs **A-H** from the list below.

Write the correct number, **i-xi**, in boxes **27-33** on your answer sheet.

List of Headings

- i Data consistency needed for language
- ii Consensus on an initiative recommendation for saving dying out languages
- iii Positive gains for protection
- iv Minimum requirement for saving a language
- v Potential threat to minority language
- vi A period when there was absent of real effort made.
- vii Native language programs launched
- viii Lack in confidence in young speakers as a negative factor
- ix Practise in several developing countries
- x Value of minority language to linguists.

xi Government participation in language field

27 Paragraph A

28 Paragraph B

29 Paragraph D

30 Paragraph E

31 Paragraph F

32 Paragraph G

33 Paragraph H

Questions 34-38

Use the information in the passage to match the people (listed **A-F**) with opinions or deeds below. Write the appropriate letters **A-F** in boxes **34-38** on your answer sheet.

A Nicholas Ostler

B Michael Krauss

C Joseph E. Grimes

D Sarah G. Thomason

E Kenneth L. Hale

F Douglas H. Whalen

34 Reported language conservation practice in Hawaii

35 Predicted that many languages would disappear soon

36 Experienced process that languages die out personally

37 Raised language fund in England

38 Not enough effort on saving until recent work

Questions 39-40

Choose the correct letter, **A, B, C** or **D**.

Write your answers in boxes 39-40 on your answer sheet.

39 What is real result of a master-apprentice program sponsored by The Ford Foundation?

A Teach children how to speak

B Revive some endangered languages in California

C Postpone the dying date for some endangered languages

D Increase communication between students

40 What should majority language speakers do according to the last paragraph?

A They should teach their children endangered language in free lessons

B They should learn at least four languages

C They should show their loyalty to a dying language

D They should be more tolerant to minority language speaker

Texting the Television

A Once upon a time, if a television show with any self-respect wanted to target a young audience, it needed to have an e-mail address. However, in Europe's TV shows, such addresses are gradually substituted by telephone numbers so that audiences can text the show from their mobile phones. Therefore, it comes as no shock that according to Gartner's research, texting has recently surpassed Internet usage across Europe. Besides, among the many uses of text messaging, one of the fastest-growing uses is to interact with television. The statistics provided by Gartner can display that 20% of French teenagers, 11% in Britain and 9% in Germany have responded to TV programmes by sending a text message.

B This phenomenon can be largely attributed to the rapid growth of reality TV shows such as 'Big Brother', where viewers get to decide the result through voting. The majority of reality shows are now open to text-message voting, and in some shows like the latest series of Norway's 'Big Brother', most votes are collected in this manner. But TV texting isn't just about voting. News shows encourage viewers to comment by texting messages; game shows enable the audience to be part of the competition; music shows answer requests by taking text messages; and broadcasters set up on-screen chatrooms. TV audiences tend to sit on the sofa with their mobile phones right by their sides, and 'it's a supernatural way to interact.' says Adam Daum of Gartner.

C Mobile service providers charge appreciable rates for messages to certain numbers, which is why TV texting can bring in a lot of cash. Take the latest British series of 'Big Brother' as an example. It brought about 5.4m text-message votes and £1.35m (\$2.1m) of profit. In Germany, MTV's 'Videoclash' encourages the audience to vote for one of two rival videos, and induces up to 40,000 texts per hour, and each one of those texts costs €0.30 (\$0.29), according to a consultancy based in Amsterdam. The Belgian quiz show '1 Against 100' had an eight-round texting match on the side, which brought in 110,000 participants in one month, and each of them paid €0.50 for each question. In Spain, a cryptic crossword clue invites the audience to send their answers through text at the expense of €1, so that they can be enrolled in the poll to win a €300 prize. Normally, 6,000 viewers would participate within one day.

At the moment, TV-related text messaging takes up a considerable proportion of mobile service providers' data revenues. In July, MmO2 (a British operator) reported an unexpectedly satisfactory result, which could be attributed to the massive text waves created by 'Big Brother'. Providers usually own 40%-50% of the profits from each text, and the rest is divided among the broadcaster, the programme producer and the company which supplies the message-processing technology. So far, revenues generated from text messages have been an indispensable part of the business model for various shows. Obviously, there has been grumbling that the providers take too much of the share. Endemol, the Netherlands-based production firm that is responsible for many reality TV shows including 'Big Brother', has begun constructing its own database for mobile phone users. It plans to set up a direct billing system with the users and bypass the providers.

D How come the joining forces of television and text message turn out to be this successful? One crucial aspect is the emergence of one-of-a-kind four-, five- or six-digit numbers known as 'shortcodes'. Every provider has control over its own shortcodes, but not until recently have they come to realise that it would make much more sense to work together to offer shortcodes compatible with all networks. The emergence of this universal short codes was a game-changer, because short codes are much easier to remember on the screen, according to Lars Becker of Flytxt, a mobile marketing company.

E Operators' cooperation on enlarging the market is by a larger trend, observes Katrina Bond of Analysys, a consultancy. When challenged by the dilemma between holding on tight to their margins and permitting the emergence of a new medium, no provider has ever chosen the latter. WAP, a technology for mobile phone users to read cut-down web pages on their screens, failed because of service providers' reluctance towards revenue-sharing with content providers. Now that they've learnt their lesson, they are altering the way of operating. Orange, a French operator, has come such a long way as to launch a rate card for sharing revenue of text messages, a new level of transparency that used to be unimaginable.

F At a recent conference, Han Weegink of CMG, a company that offers the television market text-message

infrastructure, pointed out that the television industry is changing in a subtle yet fundamental way. Instead of the traditional one-way presentation, more and more TV shows are now getting viewers' reactions involved.

Certainly, engaging the audiences more has always been the promise of interactive TV. An interactive TV was originally designed to work with exquisite set-top devices, which could be directly plugged into the TV. However, as Mr. Daum points out, that method was flawed in many ways. Developing and testing software for multiple and incompatible types of set-top box could be costly, not to mention that the 40% (or lower) market penetration is below that of mobile phones (around 85%). What's more, it's quicker to develop and set up apps for mobile phones. 'You can approach the market quicker, and you don't have to go through as many greedy middlemen,' Mr. Daum says. Providers of set-top box technology are now adding texting function to the design of their products.

G The triumph of TV-related texting reminds everyone in the business of how easily a fancy technology can all of a sudden be replaced by a less complicated, lower-tech method. That being said, the old-fashioned approach to interactive TV is not necessarily over; at least it proves that strong demands for interactive services still exist. It appears that the viewers would sincerely like to do more than simply staring at the TV screen. After all, couch potatoes would love some thumb exercises.

Questions 28 - 32

Reading Passage 3 has seven sections, **A-G**.

Choose the correct heading for sections B-E and G from the list of headings below.

Write the correct number, i-ix, in boxes **28-32** on your answer sheet.

List of Headings

- i An application of shortcodes on the TV screen
- ii An overview of a fast-growing business
- iii The trend that profitable games are gaining more concerns
- iv Why Netherlands takes the leading role
- v A new perspective towards sharing the business opportunities
- vi Factors relevant to the rapid increase in interactive TV
- vii The revenue gains and bonus share
- viii The possibility of the complex technology replaced by the simpler ones
- ix The mind change of set-top box providers

28 Section B

29 Section C

30 Section D

31 Section E

32 Section G

Questions 33 - 35

Choose the correct letter, A, B, C or D.

Write the correct letter in boxes **33-35** on your answer sheet.

33 In Europe, a research hints that young audiences spend more money on

A thumbing text messages.

B Rewriting e-mails.

C watching TV programmes.

D talking through mobile phones.

34 What would happen when reality TV shows invite the audience to vote?

A Viewers would get attractive bonus.

- B They would be part of the competition.
- C Their questions would be replied.
- D Their participation could change the result.

35 Interactive TV will change from concentrating on set-top devices to

- A increasing their share in the market.
- B setting up a modified set-top box.
- C building an embedded message platform.
- D marching into the European market.

Questions 36 - 40

Look at the following descriptions (Questions **36-40**) and the list of companies below.

Match each description with the correct company, **A-F**.

Write the correct letter, **A-F**, in boxes **36-40** on your answer sheet.

List of Companies

- A Flytxt
- B Analysys
- C Endemol
- D CMG
- E MmO2
- F Gartner

36 offered mobile phone message technology

37 earned considerable amount of money through a famous programme

38 expressed the view that shortcodes are convenient to remember when turning up

39 built their own mobile phone operating applications

40 indicated that it is easy for people to send message in an interactive TV

What Do Babies Know?

As Daniel Haworth is settled into a high chair and wheeled behind a black screen, a sudden look of worry furrows his 9-month-old brow. His dark blue eyes dart left and right in search of the familiar reassurance of his mother's face. She calls his name and makes soothing noises, but Daniel senses something unusual is happening. He sucks his fingers for comfort, but, finding no solace, his mouth crumples, his body stiffens, and he lets rip an almighty shriek of distress. This is the usual expression when babies are left alone or abandoned. Mom picks him up, and reassures him, and two minutes later, a chortling and alert Daniel returns to the darkened booth behind the screen and submits himself to baby lab, a unit set up in 2005 at the University of Manchester in northwest England to investigate how babies think.

Watching infants piece life together, seeing their senses, emotions and motor skills take shape, is a source of mystery and endless fascination—at least to parents and developmental psychologists. We can decode their signals of distress or read a million messages into their first smile. But how much do we really know about what's going on behind those wide, innocent eyes? How much of their understanding of and response to the world comes preloaded at birth? How much is built from scratch by experience? Such are the questions being explored at baby lab. Though the facility is just 18 months old and has tested only 100 infants, it's already challenging current thinking on what babies know and how they come to know it.

Daniel is now engrossed in watching video clips of a red toy train on a circular track. The train disappears into a tunnel and emerges on the other side. A hidden device above the screen is tracking Daniel's eyes as they follow the train and measuring the diameter of his pupils 50 times a second. As the child gets bored—or "habituated", as psychologists call the process—his attention level steadily drops. But it picks up a little whenever some novelty is introduced. The train might be green, or it might be blue. And sometimes an impossible thing happens—the train goes into the tunnel one color and comes out another.

Variations of experiments like this one, examining infant attention, have been a standard tool of developmental psychology ever since the Swiss pioneer of the field, Jean Piaget, started experimenting on his children in the 1920s. Piaget's work led him to conclude that infants younger than 9 months have no innate knowledge of how the world works or any sense of "object permanence" (that people and things still exist even when they're not seen). Instead, babies must gradually construct this knowledge from experience. Piaget's "constructivist" theories were massively influential on postwar educators and psychologists, but over the past 20 years or so they have been largely set aside by a new generation of "nativist" psychologists and cognitive scientists whose more sophisticated experiments led them to theorise that infants arrive already equipped with some knowledge of the physical world and even rudimentary programming for math and language. Baby lab director Sylvain Sirois has been putting these smart-baby theories through a rigorous set of tests. His conclusions so far tend to be more Piagetian: "Babies," he says, "know nothing."

What Sirois and his postgraduate assistant Lain Jackson are challenging is the interpretation of a variety of classic experiments begun in the mid-1980s in which babies were shown physical events that appeared to violate such basic concepts as gravity, solidity and contiguity. In one such experiment, by University of Illinois psychologist Renee Baillargeon, a hinged wooden panel appeared to pass right through a box. Baillargeon and M.I.T.'s Elizabeth Spelke found that babies as young as 3 1/2 months would reliably look longer at the impossible event than at the normal one, their conclusion: babies have enough built-in knowledge to recognise that something is wrong.

Sirois does not take issue with the way these experiments were conducted. "The methods are correct and replicable," he says, "it's the interpretation that's the problem." In a critical review to be published in the forthcoming issue of the *European Journal of Developmental Psychology*, he and Jackson pour cold water over recent experiments that claim to have observed innate or precocious social cognition skills in infants. His own experiments indicate that a baby's fascination with physically impossible events merely reflects a response to stimuli that are novel. Data from the eye tracker and the measurement of the pupils (which widen in response to arousal or interest) show that impossible events involving familiar objects are no more interesting than possible events involving novel objects. In other words, when Daniel had seen the red train come out of the tunnel green a few times, he gets as bored as when it stays the same color. The mistake of previous research, says Sirois, has been to leap to the conclusion that infants can understand the concept of impossibility from the mere fact that they are able to perceive some novelty in it. "The real explanation is boring," he says.

So how do babies bridge the gap between knowing squat and drawing triangles—a task Daniel's sister Lois, 2 1/2, is happily tackling as she waits for her brother? "Babies have to learn everything, but as Piaget was saying, they start with a few primitive reflexes that get things going," said Sirois. For example, hardwired in the brain is an instinct that draws a baby's eyes to a human face. From brain imaging studies we also know that the brain has some sort of visual buffer that continues to represent objects after they have been removed—a lingering perception rather than conceptual understanding. So when babies encounter novel or unexpected events, Sirois explains, "there's a mismatch between the buffer and the information they're getting at that moment. And what you do when you've

got a mismatch is you try to clear the buffer. And that takes attention." So learning, says Sirois, is essentially the laborious business of resolving mismatches. "The thing is, you can do a lot of it with this wet sticky thing called a brain. It's a fantastic, statistical-learning machine". Daniel, exams ended, picks up a plastic tiger and, chewing thoughtfully upon its heat, smiles as if to agree.

Questions 27-32

Do the following statements agree with the information given in Reading Passage 3?

In boxes **27-32** on your answer sheet, write

- TRUE if the statement agrees with the information
FALSE if the statement contradicts with the information
NOT GIVEN if there is no information on this

- 27** Baby's behavior after being abandoned is not surprising.
28 Parents are overestimating what babies know.
29 Only 100 experiments have been done but can prove the theories about what we know.
30 Piaget's theory was rejected by parents in 1920s.
31 Sylvain Sirois's conclusion on infant's cognition is similar to Piaget's.
32 Sylvain Sirois found serious flaws in the experimental designs by Baillargeon and Elizabeth Spelke.

Questions 33-37

Complete each sentence with the correct ending, **A-E**, below.

Write the correct letter, **A-E**, in boxes **33-37** on your answer sheet.

- 33** Jean Piaget thinks infants younger than 9 months won't know something existing
34 Jean Piaget thinks babies only get the knowledge
35 Some cognitive scientists think babies have the mechanism to learn a language
36 Sylvain Sirois thinks that babies can reflect a response to stimuli that are novel
37 Sylvain Sirois thinks babies' attention level will drop
A before they are born.
B before they learn from experience.
C when they had seen the same thing for a while.
D when facing the possible and impossible events.
E when the previous things appear again in the lives.

Questions 38-40

Choose the correct letter, A, B, C or D.

Write the correct letter in boxes 38-40 on your answer sheet.

- 38** What can we know about Daniel in the third paragraph?
A Daniel's attention level rose when he saw a blue train.
B Kid's attention fell when he was accustomed to the changes.
C Child's brain activity was monitored by a special equipment.
D Size of the train changed when it came out of the tunnel.
39 What can we know from the writer in the fourth paragraph?
A The theories about what baby knows changed over time.
B Why the experiments that had been done before were rejected.
C Infants have the innate knowledge to know the external environment.
D Piaget's "constructivist" theories were massively influential on parents.
40 What can we know from the argument of the experiment about the baby in the sixth paragraph?
A Infants are attracted by various colours of the trains all the time.

- B Sylvain Sirois accuses misleading approaches of current experiments.
- C Sylvain Sirois indicates that only impossible events make children interested.
- D Sylvain Sirois suggests that novel things attract baby's attention.

The Rainmaker Design

A Sometimes ideas just pop up out of the blue. Or in Charlie Paton's case, out of the rain. 'I was in a bus in Morocco traveling through the desert,' he remembers. 'It had been raining and the bus was full of hot, wet people. The windows steamed up and I went to sleep with a towel against the glass. When I woke, the thing was soaking wet. I had to wring it out. And it set me thinking. Why was it so wet?'

B The answer, of course, was condensation. Back home in London, a physicist friend, Philip Davies, explained that the glass, chilled by the rain outside, had cooled the hot humid air inside the bus below its dew point, causing droplets of water to form on the inside of the window. Intrigued, Paton—a lighting engineer by profession—started rigging up his own equipment. 'I made my own solar stills. It occurred to me that you might be able to produce water in this way in the desert, simply by cooling the air. I wondered whether you could make enough to irrigate fields and grow crops.'



C Today, a decade on, his dream has taken shape as a giant greenhouse on a desert island off Abu Dhabi in the Persian Gulf—the first commercially viable version of his 'seawater greenhouse'. Local scientists, working with Paton, are watering the desert and growing vegetables in what is basically a giant dew-making machine that produces fresh water and cool air from sun and seawater. In awarding Paton first prize in a design competition two years ago, Marco Goldschmied, president of the Royal Institute of British Architects, called it 'a truly original idea which has the potential to impact on the lives of millions of people living in coastal water-starved areas around the world'.

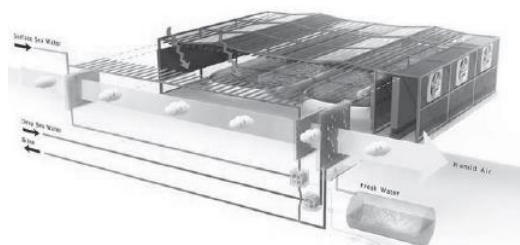


D The seawater greenhouse as developed by Paton has three main parts. They both air-condition the greenhouse and provide water for irrigation. The front of the greenhouse faces into the prevailing wind so that hot dry air blows in through a front wall. The wall made of perforated cardboard kept moist by a constant trickle of seawater pumped up from ocean. The purpose is to cool and moisten the incoming desert air. The cool moist air allows the plants to grow faster. And, crucially, because much less water evaporates from the leaves, the plants need much less

moisture to grow than if they were being irrigated in the hot dry desert air outside the greenhouse.

E The air-conditioning of the interior of the greenhouse is completed by the second feature: the roof. It has two layers: an outer layer of clear polyethylene and an inner coated layer that reflects infrared radiation. This combination ensures that visible light can stream through to the plants, maximizing the rate of plant growth through photosynthesis but at the same time heat from the infrared radiation is trapped in the space between the layers, and kept away from the plants. This helps keep the air around the plants cool.

F At the back of the greenhouse sits the third elements. This is the main water production unit. Here, the air hits a second moist cardboard wall that increases its humidity as it reaches the condenser, which finally collects from the hot humid air the moisture for irrigating the plants. The condenser is a metal surface kept cool by stilling more seawater. It is the equivalent of the window on Paton's Moroccan bus. Drops of pure distilled water form on the condenser and flow into a tank for irrigating the crops.



G The Abu Dhabi greenhouse more or less runs itself. Sensors switch everything on when the sun rises and alter flows of air and seawater through the day in response to changes in temperature, humidity, and sunlight. On windless days, fans ensure a constant flow of air through the greenhouse. 'Once it is tuned to the local environment, you don't need anyone there for it to work,' says Paton. 'We can run the entire operation off one 13-amp plug, and in the future we could make it entirely independent of the grid, powered from a few solar panels.'

H Critics point out that construction costs of around \$4 a square foot are quite high. By illustration, however, Paton presents that it can cool as efficiently as a 500-kilowatt air conditioner while using less than 3 kilowatts of electricity. Thus the plants need only an eighth of the volume of water used by those grown conventionally. And so the effective cost of the desalinated water in the greenhouse is only a quarter that of water from a standard desalinator, which is good economics. Besides it really suggests an environmentally-friendly way of providing air conditioning on a scale large enough to cool large greenhouses where crops can be grown despite the high outside temperatures.

Questions 27-31

Do the following statements agree with the claims of the writer in Reading Passage 3?

In boxes **27-31** on your answer sheet, write

YES if the statement is true

NO if the statement is false

NOT GIVEN if the information is not given in the passage

27 The idea just came to Charlie Paton by accident.

28 The bus was well ventilated.

29 After waking up, Paton found his towel was wet.

30 The fan in the bus did not work well.

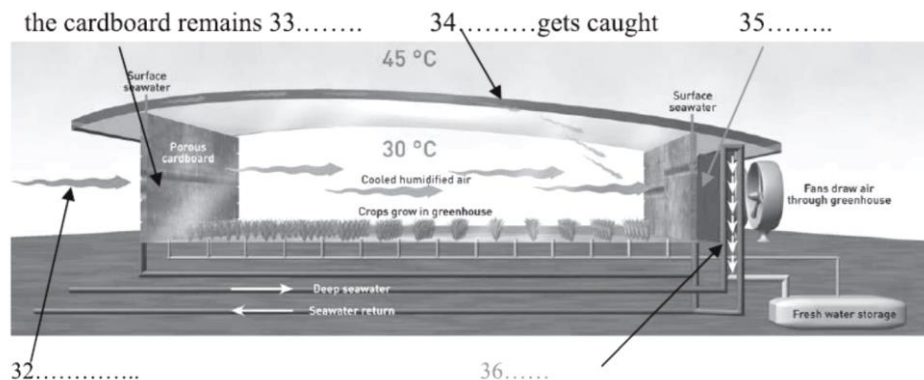
31 Paton immediately operated his own business in Persian Gulf after talking with Philip Davies.

Questions 32-36

Label the diagram below.

Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

Write your answers in boxes **32-36** on our answer sheet.



Questions 37-40

Summary

Complete the summary below, using **NO MORE THAN TWO WORDS** from the passage for each answer.

Write your answers in boxes **37-40** on your answer sheet.

To some extent, the Abu Dhabi greenhouse functions automatically. When the day is sunny, the equipment can respond to the changes in several natural elements. When there is no wind, **37** _____ help to retain the flow of air. Even in the future, we have an ideal plan to power the greenhouse from **38** _____. However, there are still some critics who argue that **39** _____ are not good economics. To justify himself, Paton presents favorable arguments against these critics and suggests that it is an **40** _____ approach to provide air conditioning in a scale large sense.

Answer keys

Passage1

Satellite Technology

1. iv
2. vii
3. iii
- 4 ii
5. ix
6. F
7. B
8. D
9. A
10. FALSE
11. NOT GIVEN
12. TRUE
13. TRUE

Mechanisms of Linguist Change

- 1 sound laws
- 2 fashion
- 3 imperfect
- 4 principle of ease
- 5 FALSE
- 6 FALSE
- 7 NOT GIVEN
- 8 TRUE
- 9 TRUE
- 10 NOT GIVEN
- 11 TRUE
- 12 C
- 13 B
- 14 A

Animal Minds: Parrot Alex

1. NOT GIVEN
2. NOT GIVEN
3. FALSE
4. TRUE
5. TRUE
6. FALSE
7. 100 English words
8. chimpanzees
9. avian cognition

10. particularly chosen
11. color
12. wrong pronunciation
13. teenager

The Impact of the Potato

- 1.FALSE
- 2.FALSE
- 3.NOT GIVEN
- 4.TRUE
- 5.TRUE
- 6.flower
- 7.prejudice
- 8.reverse
- 9.meat
- 10.crops
- 11.soil
- 12.cultivation
- 13.investment

Ancient Storytelling

1. E
2. G
3. A
4. B
5. H
6. B
7. B
8. C
9. A
10. the Poetics
11. tragedy
12. landmarks
13. flaw/weakness

William Gilbert and Magnetism

1. v
2. i
3. vi
4. x
5. ix
6. iv
7. ii
8. TRUE

- 9. TRUE
- 10. NOT GIVEN
- 11. C
- 12. D
- 13. E

Man or Machine

- 1 E
- 2 C
- 3 F
- 4 A
- 5 F
- 6 B
- 7 17 years
- 8 backpack
- 9 interact with
- 10 facial expressions
- 11 Cog/Cognition
- 12 sensors
- 13 intelligence

The Innovation of Grocery Stores

- 1 D
- 2 A
- 3 F
- 4 C
- 5 E
- 6 clerk
- 7 customers
- 8 lobby
- 9 stockroom
- 10 galleries
- 11-13 CBC

The Research for Intelligence

- 1 v
- 2 i
- 3 ii
- 4 iii
- 5 vii
- 6 x
- 7 E
- 8 A
- 9 F

- 10 D
- 11 micro-arrays
- 12 intricate dance steps
- 13 a reflection

Biomimetic Design

- 1 NOT GIVEN
- 2 FALSE
- 3 TRUE
- 4 FALSE
- 5 NOT GIVEN
- 6 FALSE
- 7 TRUE
- 8 the same way
- 9 carbon-fiber
- 10 limbs
- 11 self-cleaning
- 12 surveillance
- 13 lifesaving

Passage 2

Novice and Expert-Becoming an Expert

- 14. guiding principles and rules
- 15. mentor
- 16. journeyman
- 17. patterns of behavior
- 18. complex
- 19. FALSE
- 20. NOT GIVEN
- 21. TRUE
- 22. FALSE
- 23. TRUE
- 24. models
- 25. human bias
- 26. consensus

Movie of Metropolis

- 14. YES
- 15. NOT GIVEN
- 16. NO
- 17. NOT GIVEN
- 18. social division
- 19. machines

- 20. John Frederse
- 21. abstract
- 22. function
- 23. efficiency
- 24. C
- 25. A
- 26. B
- 27. D

Communicating Conflict

- 14. iii
- 15. vii
- 16. i
- 17. iv
- 18. ix
- 19. viii
- 20. v
- 21. ii
- 22. FALSE
- 23. TRUE
- 24. NOT GIVEN
- 25. TRUE
- 26. TRUE
- 27. B

Memory and Age

- 14. C
- 15. D
- 16. B
- 17. C
- 18. Memory
- 19. psychological
- 20. semantic memory
- 21. episodic memory/event memory
- 22. algebra
- 23. vocabulary
- 24. E
- 25. B
- 26. A
- 27. C

Seed Hunting

- 14. TRUE
- 15. NOT GIVEN

- 16. TRUE
- 17. TRUE
- 18. FALSE
- 19. TRUE
- 20. extinction
- 21. drugs, crops
- 22. pioneers
- 23. Sir Joseph Banks
- 24. underground vaults
- 25. A
- 26. B

Can Scientists Tell Us: What happiness is?

- 14 B
- 15 A
- 16 F
- 17 C
- 18 G
- 19 H
- 20 E
- 21 candy
- 22 definition
- 23 catastrophic brain
- 24 landscapes or dolphins playing
- 25 (more) primitive parts
- 26 D

Bird Migration

- 14 iv
- 15 v
- 16 ii
- 17 x
- 18 vii
- 19 i
- 20 viii
- 21 A
- 22 C
- 23 parental guidance
- 24 compass
- 25 predators
- 26 visible

Facial Expression

- 14 D

15 A
16 C
17 H
18 D
19 B
20 misidentified
21 emotions
22 cultural background
23 isolated
24 exposed
25-26 CE

Leaf-Cutting Ants and Fungus

14 B
15 A
16 A
17 C
18 B
19 A
20-24 FHCJG
25-26 AC

Left-handed or Right-handed

14 C
15 A
16 B
17 F
18 D
19 D
20 B
21 A
22 C
23 YES
24 NOT GIVEN
25 NO
26 NOT GIVEN

Passage 3

Learning to Walk

27 TRUE
28 FALSE
29 TRUE
30 TRUE
31 NOT GIVEN

32 A
33 C
34 D
35 anatomy
36 resistance
37 stress
38 hypertension
39 organs
40 soles

Mind Music

27 A
28 E
29 D
29 H
30 wrist
31 emotions
32 unoccupied
33 challenging
34 C
35 A
36 B
37 C
38 B

The role of accidents in business

27 YES
28 NO
29 NO
30 YES
31 NOT GIVEN
32 F
33 G
34 C
35 A
36 B
37 D
38 A
39 C
40 B

Global Warming in New Zealand

27 D
28 B

- 29 A
- 30 C
- 31 A
- 32 A
- 33 high tides
- 34 agricultural production
- 35 coastal boundaries
- 36 NOT GIVEN
- 37 NOT GIVEN
- 38 NO
- 39 YES
- 40 NO

The Significant Role of Mother Tongue in Education

- 27 C
- 28 A
- 29 B
- 30 D
- 31 I
- 32 D
- 33 J
- 34 F
- 35 C
- 36 YES
- 37 NOT GIVEN
- 38 NO
- 39 NOT GIVEN
- 40 YES

What Do Babies Know?

- 27 Ture
- 28 Not Given
- 29 False
- 30 Not Given
- 31 Ture
- 32 False
- 33 B
- 34 E
- 35 A
- 36 D
- 37 C
- 38 B
- 39 A
- 40 D

Sand Dunes

27 i
28 v
29 x
30 vii
31 ix
32 ii
33 vi
34 iv
35-36 BC
37 B
38 G
39 H
40 D

Save Endangered Language

27 v
28 x
29 iii
30 i
31 vii
32 viii
33 ii
34 C
35 B
36 E
37 A
38 D
39 C
40 D

The Rainmaker Design

27 YES
28 NO
29 YES
30 NOT GIVEN
31 NO
32 hot dry air
33 moist
34 heat
35 pure distilled waster
36 condenser
37 fans
38 solar panels

39 construction costs
40 environmentally-friendly

Texting the Television

28 vi
29 vii
30 i
31 v
32 viii
33 A
34 D
35 C
36 D
37 E
38 A
39 C
40 F